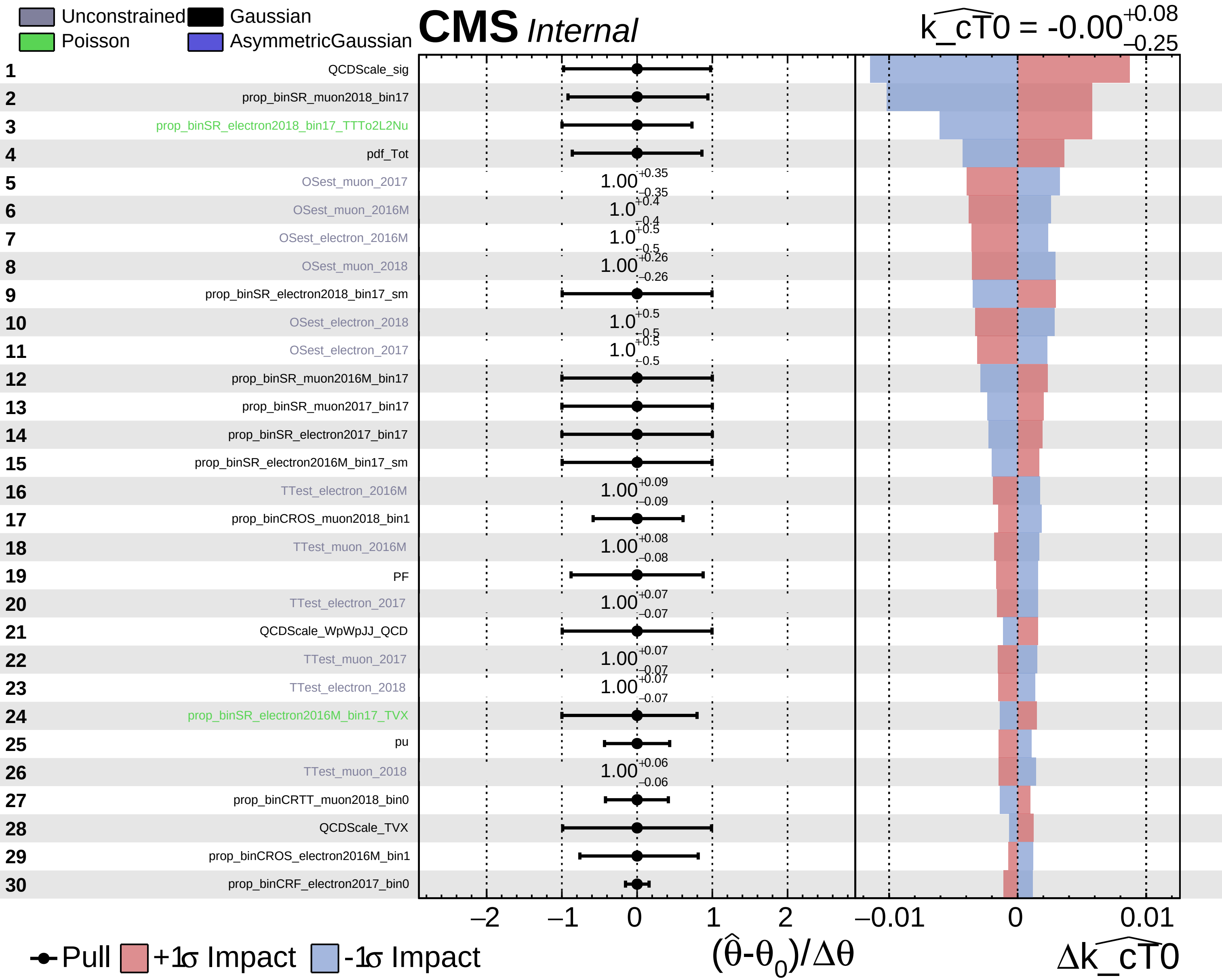
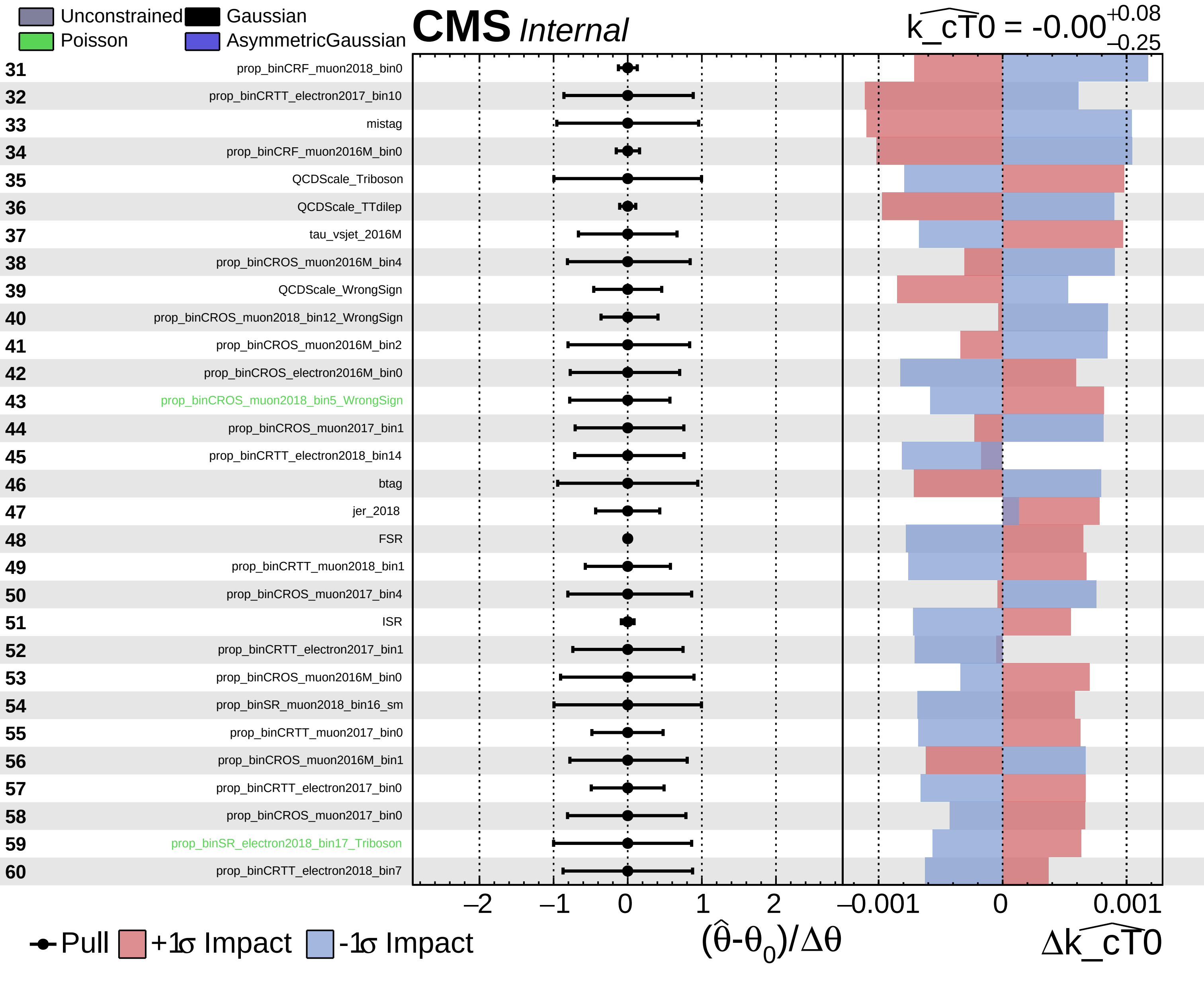
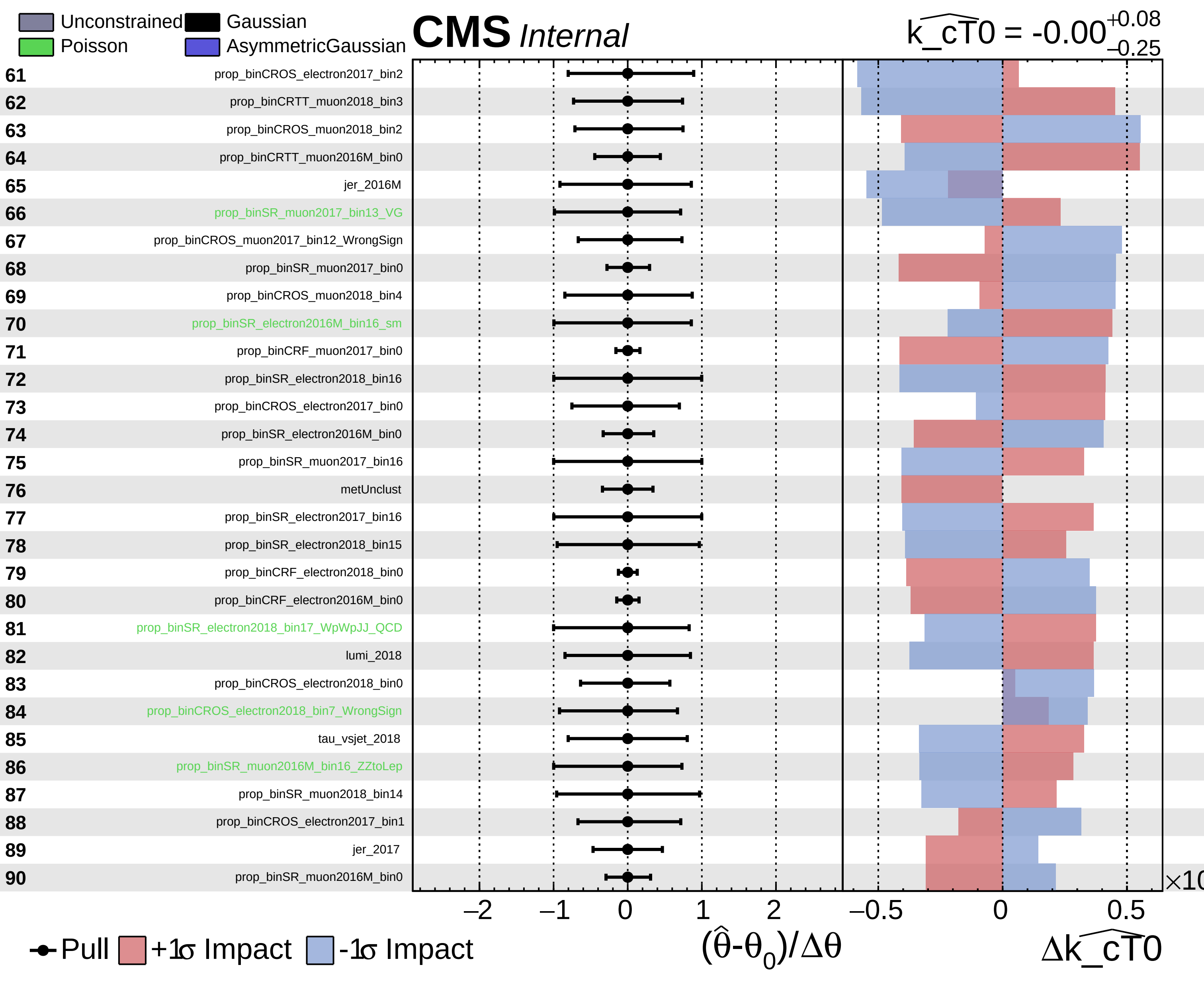
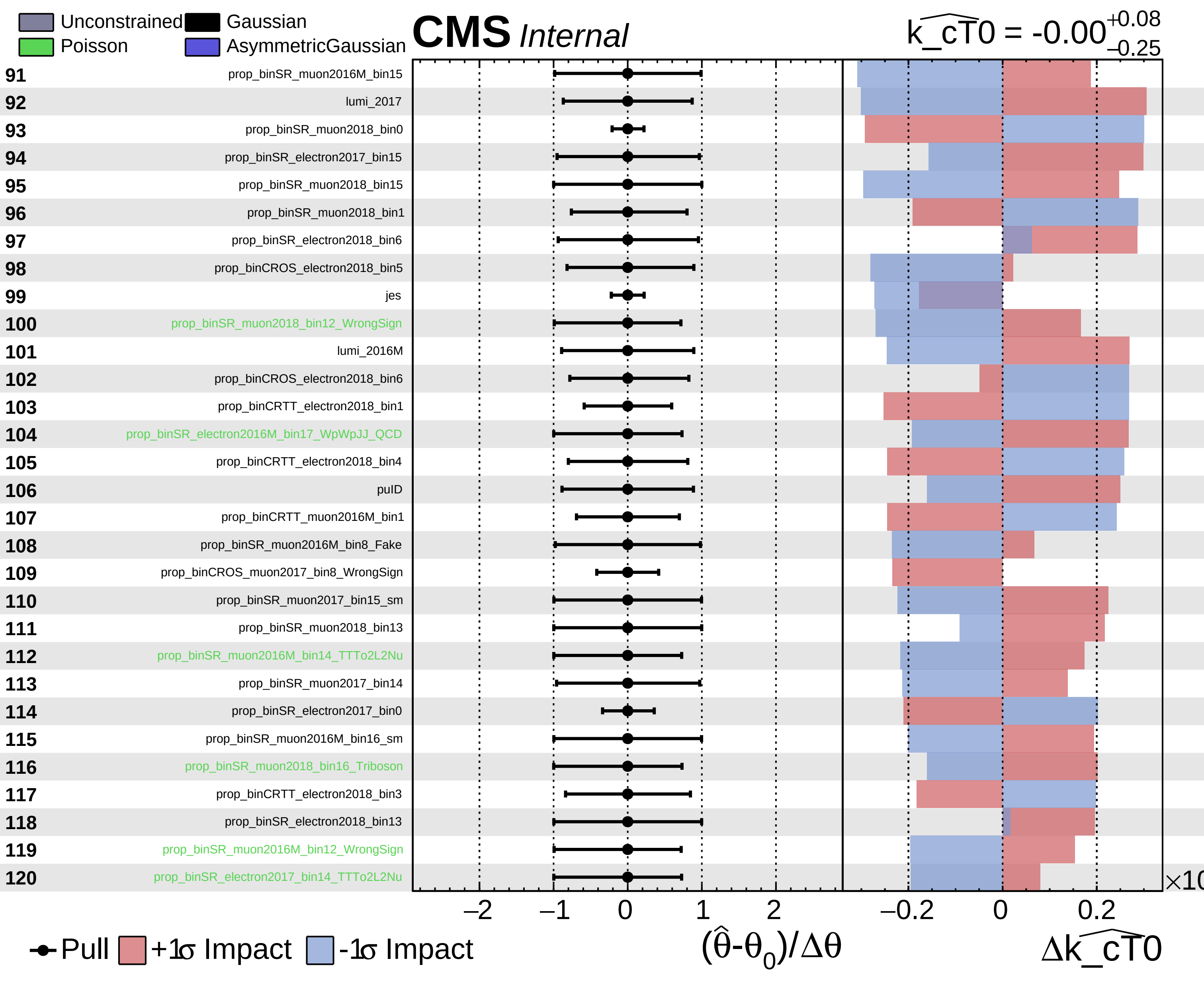


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
1 QCDScale_sig
2 prop_binSR_muon2018_bin17
3 prop_binSR_electron2018_bin17_TTTo2L2Nu
4 pdf_Tot
5 OSest_muon_2017
6 OSest_muon_2016M
7 OSest_electron_2016M
8 OSest_muon_2018
9 prop_binSR_electron2018_bin17_sm
10 OSest_electron_2018
11 OSest_electron_2017
12 prop_binSR_muon2016M_bin17
13 prop_binSR_muon2017_bin17
14 prop_binSR_electron2017_bin17
15 prop_binSR_electron2016M_bin17_sm
16 TTest_electron_2016M
17 prop_binCROS_muon2018_bin1
18 TTest_muon_2016M
19 PF
20 TTest_electron_2017
21 QCDScale_WpWpJJ_QCD
22 TTest_muon_2017
23 TTest_electron_2018
24 prop_binSR_electron2016M_bin17_TVX
25 pu
26 TTest_muon_2018
27 prop_binCRTT_muon2018_bin0
28 QCDScale_TVX
29 prop_binCROS_electron2016M_bin1
30 prop_binCRF_electron2017_bin0
1.00 +0.35 −0.35
1.0 +0.4 −0.4
1.0 +0.5 −0.5
1.00 +0.26 −0.26
1.0 +0.5 −0.5
1.0 +0.5 −0.5
1.00 +0.09 −0.09
1.00 +0.08 −0.08
1.00 +0.07 −0.07
1.00 +0.07 −0.07
1.00 +0.07 −0.07
1.00 +0.06 −0.06
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0

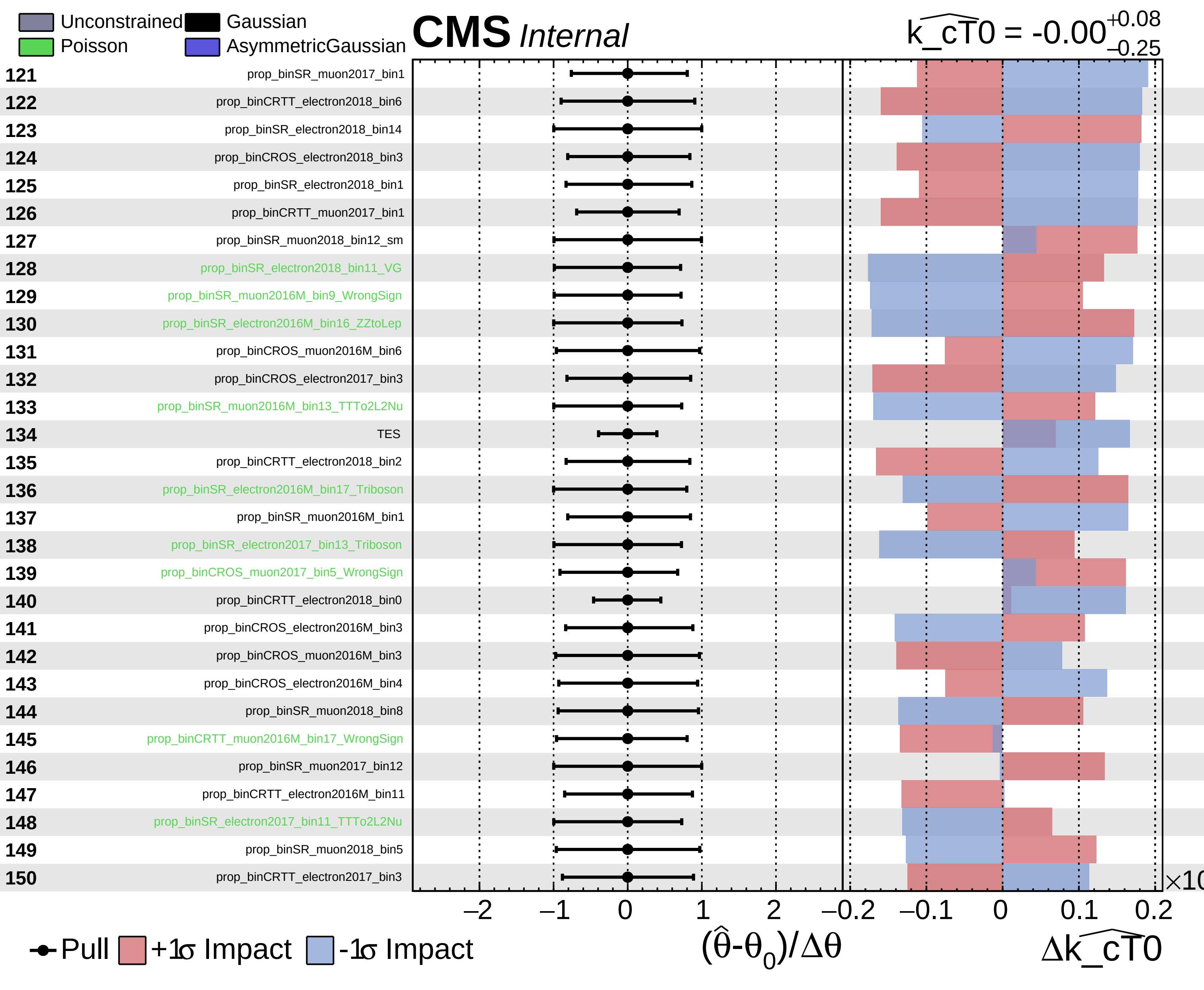


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
31 prop_binCRF_muon2018_bin0
32 prop_binCRTT_electron2017_bin10
33 mistag
34 prop_binCRF_muon2016M_bin0
35 QCDScale_Triboson
36 QCDScale_TTdilep
37 tau_vsjet_2016M
38 prop_binCROS_muon2016M_bin4
39 QCDScale_WrongSign
40 prop_binCROS_muon2018_bin12_WrongSign
41 prop_binCROS_muon2016M_bin2
42 prop_binCROS_electron2016M_bin0
43 prop_binCROS_muon2018_bin5_WrongSign
44 prop_binCROS_muon2017_bin1
45 prop_binCRTT_electron2018_bin14
46 btag
47 jer_2018
48 FSR
49 prop_binCRTT_muon2018_bin1
50 prop_binCROS_muon2017_bin4
51 ISR
52 prop_binCRTT_electron2017_bin1
53 prop_binCROS_muon2016M_bin0
54 prop_binSR_muon2018_bin16_sm
55 prop_binCRTT_muon2017_bin0
56 prop_binCROS_muon2016M_bin1
57 prop_binCRTT_electron2017_bin0
58 prop_binCROS_muon2017_bin0
59 prop_binSR_electron2018_bin17_Triboson
60 prop_binCRTT_electron2018_bin7
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

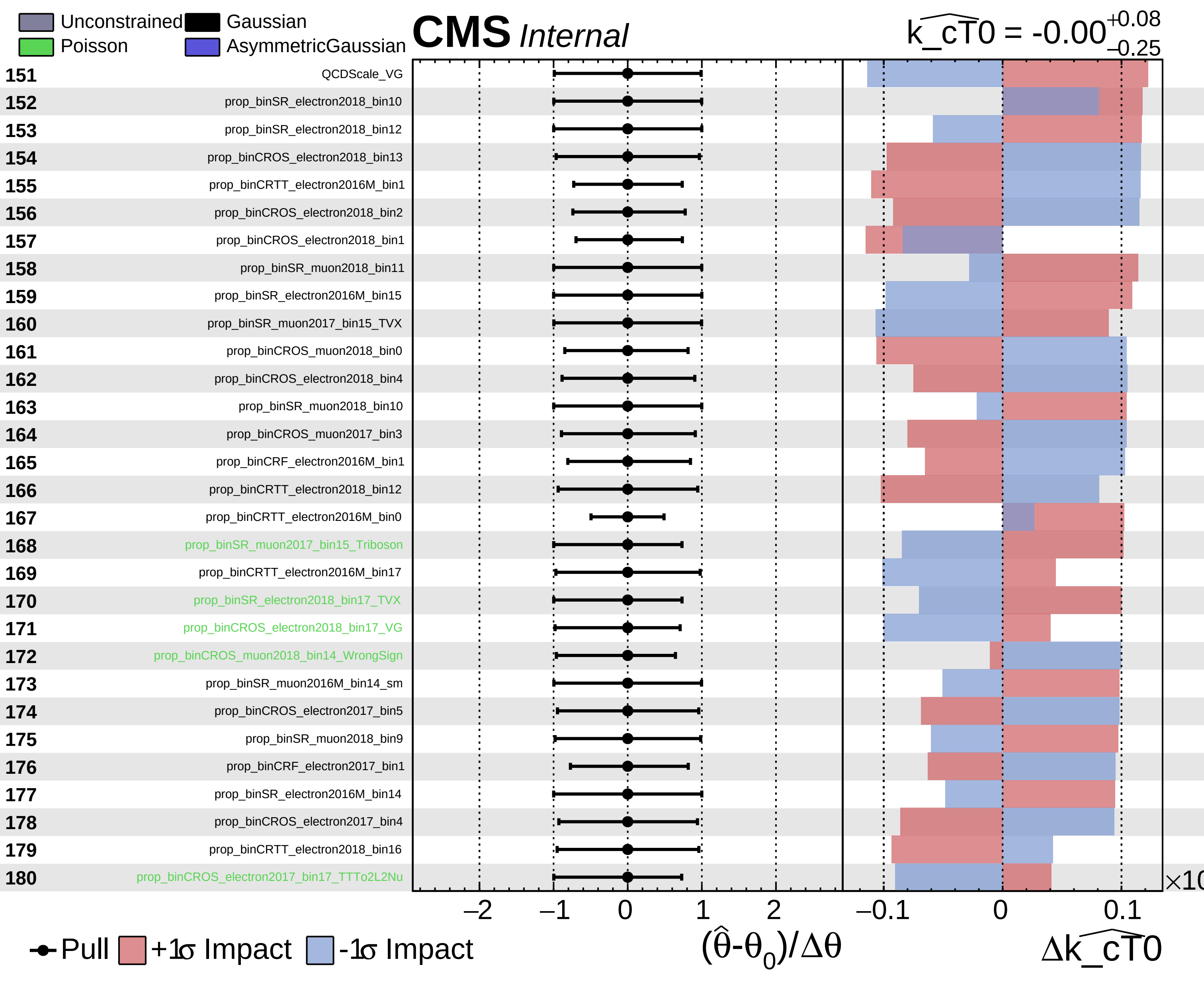

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
61 prop_binCROS_electron2017_bin2
62 prop_binCRTT_muon2018_bin3
63 prop_binCROS_muon2018_bin2
64 prop_binCRTT_muon2016M_bin0
65 jer_2016M
66 prop_binSR_muon2017_bin13_VG
67 prop_binCROS_muon2017_bin12_WrongSign
68 prop_binSR_muon2017_bin0
69 prop_binCROS_muon2018_bin4
70 prop_binSR_electron2016M_bin16_sm
71 prop_binCRF_muon2017_bin0
72 prop_binSR_electron2018_bin16
73 prop_binCROS_electron2017_bin0
74 prop_binSR_electron2016M_bin0
75 prop_binSR_muon2017_bin16
76 metUnclust
77 prop_binSR_electron2017_bin16
78 prop_binSR_electron2018_bin15
79 prop_binCRF_electron2018_bin0
80 prop_binCRF_electron2016M_bin0
81 prop_binSR_electron2018_bin17_WpWpJJ_QCD
82 lumi_2018
83 prop_binCROS_electron2018_bin0
84 prop_binCROS_electron2018_bin7_WrongSign
85 tau_vsjet_2018
86 prop_binSR_muon2016M_bin16_ZZtoLep
87 prop_binSR_muon2018_bin14
88 prop_binCROS_electron2017_bin1
89 jer_2017
90 prop_binSR_muon2016M_bin0
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



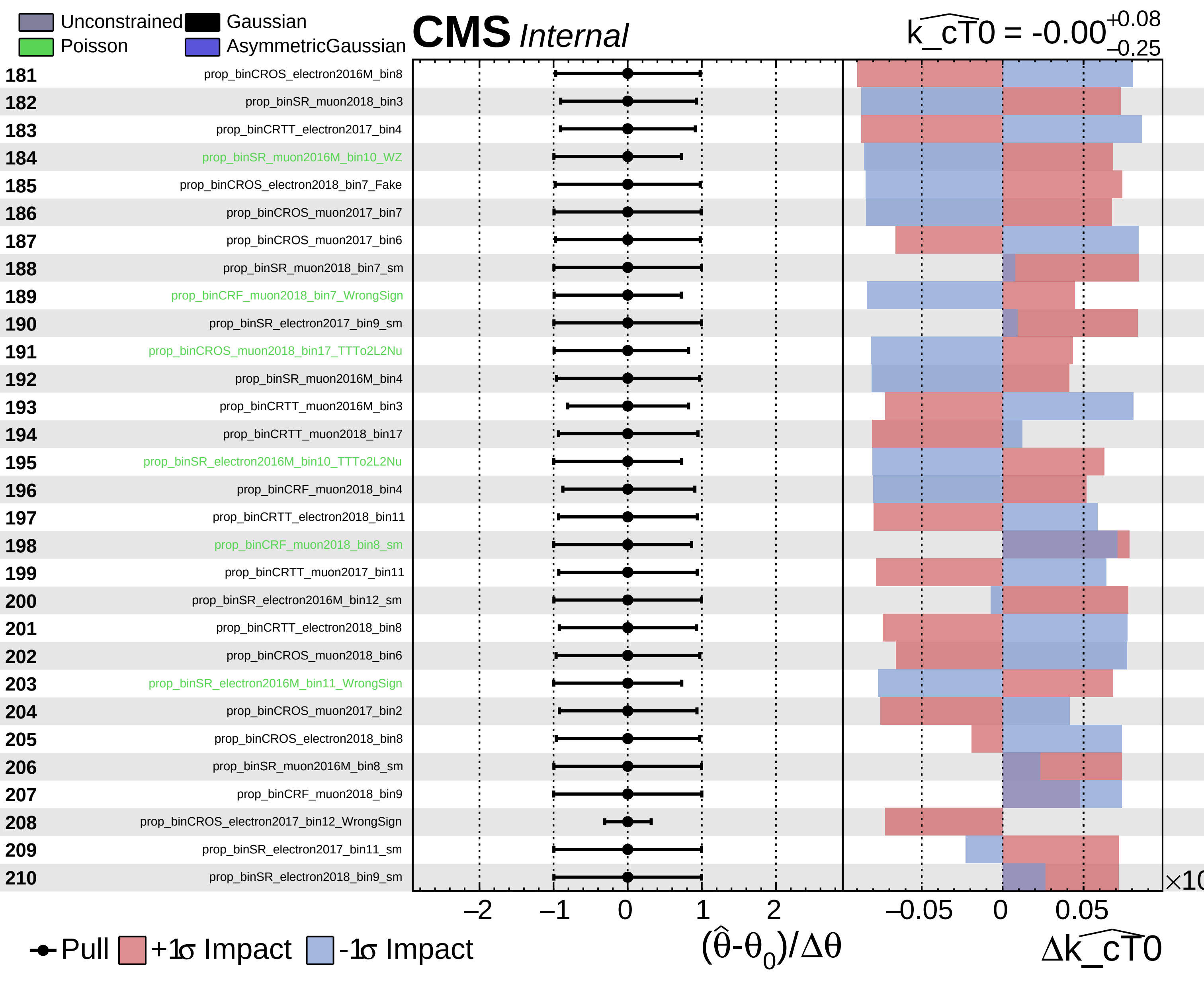
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
91 prop_binSR_muon2016M_bin15
92 lumi_2017
93 prop_binSR_muon2018_bin0
94 prop_binSR_electron2017_bin15
95 prop_binSR_muon2018_bin15
96 prop_binSR_muon2018_bin1
97 prop_binSR_electron2018_bin6
98 prop_binCROS_electron2018_bin5
99 jes
100 prop_binSR_muon2018_bin12_WrongSign
101 lumi_2016M
102 prop_binCROS_electron2018_bin6
103 prop_binCRTT_electron2018_bin1
104 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
105 prop_binCRTT_electron2018_bin4
106 puID
107 prop_binCRTT_muon2016M_bin1
108 prop_binSR_muon2016M_bin8_Fake
109 prop_binCROS_muon2017_bin8_WrongSign
110 prop_binSR_muon2017_bin15_sm
111 prop_binSR_muon2018_bin13
112 prop_binSR_muon2016M_bin14_TTTo2L2Nu
113 prop_binSR_muon2017_bin14
114 prop_binSR_electron2017_bin0
115 prop_binSR_muon2016M_bin16_sm
116 prop_binSR_muon2018_bin16_Triboson
117 prop_binCRTT_electron2018_bin3
118 prop_binSR_electron2018_bin13
119 prop_binSR_muon2016M_bin12_WrongSign
120 prop_binSR_electron2017_bin14_TTTo2L2Nu
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



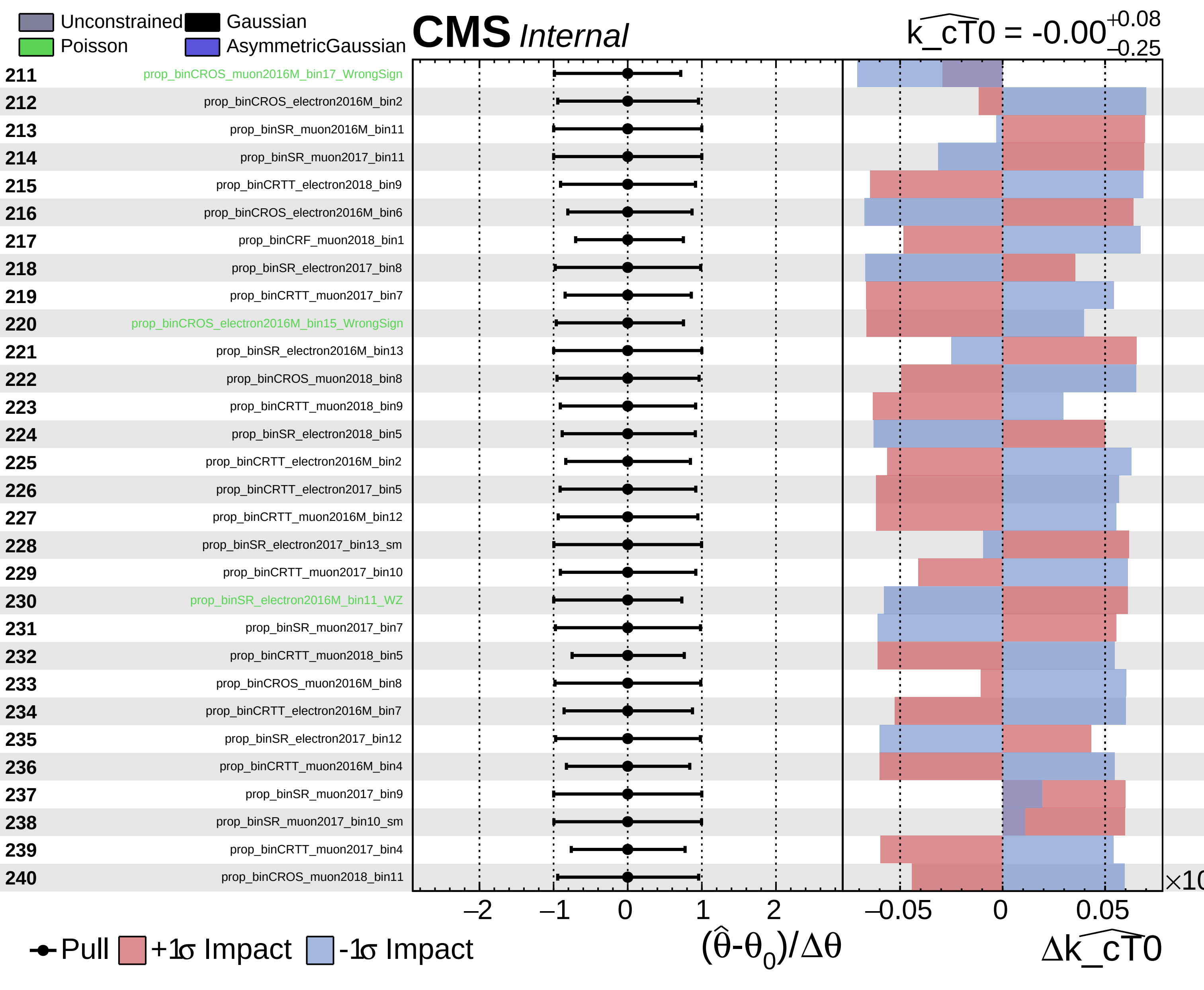
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
121 prop_binSR_muon2017_bin1
122 prop_binCRTT_electron2018_bin6
123 prop_binSR_electron2018_bin14
124 prop_binCROS_electron2018_bin3
125 prop_binSR_electron2018_bin1
126 prop_binCRTT_muon2017_bin1
127 prop_binSR_muon2018_bin12_sm
128 prop_binSR_electron2018_bin11_VG
129 prop_binSR_muon2016M_bin9_WrongSign
130 prop_binSR_electron2016M_bin16_ZZtoLep
131 prop_binCROS_muon2016M_bin6
132 prop_binCROS_electron2017_bin3
133 prop_binSR_muon2016M_bin13_TTTo2L2Nu
134 TES
135 prop_binCRTT_electron2018_bin2
136 prop_binSR_electron2016M_bin17_Triboson
137 prop_binSR_muon2016M_bin1
138 prop_binSR_electron2017_bin13_Triboson
139 prop_binCROS_muon2017_bin5_WrongSign
140 prop_binCRTT_electron2018_bin0
141 prop_binCROS_electron2016M_bin3
142 prop_binCROS_muon2016M_bin3
143 prop_binCROS_electron2016M_bin4
144 prop_binSR_muon2018_bin8
145 prop_binCRTT_muon2016M_bin17_WrongSign
146 prop_binSR_muon2017_bin12
147 prop_binCRTT_electron2016M_bin11
148 prop_binSR_electron2017_bin11_TTTo2L2Nu
149 prop_binSR_muon2018_bin5
150 prop_binCRTT_electron2017_bin3
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



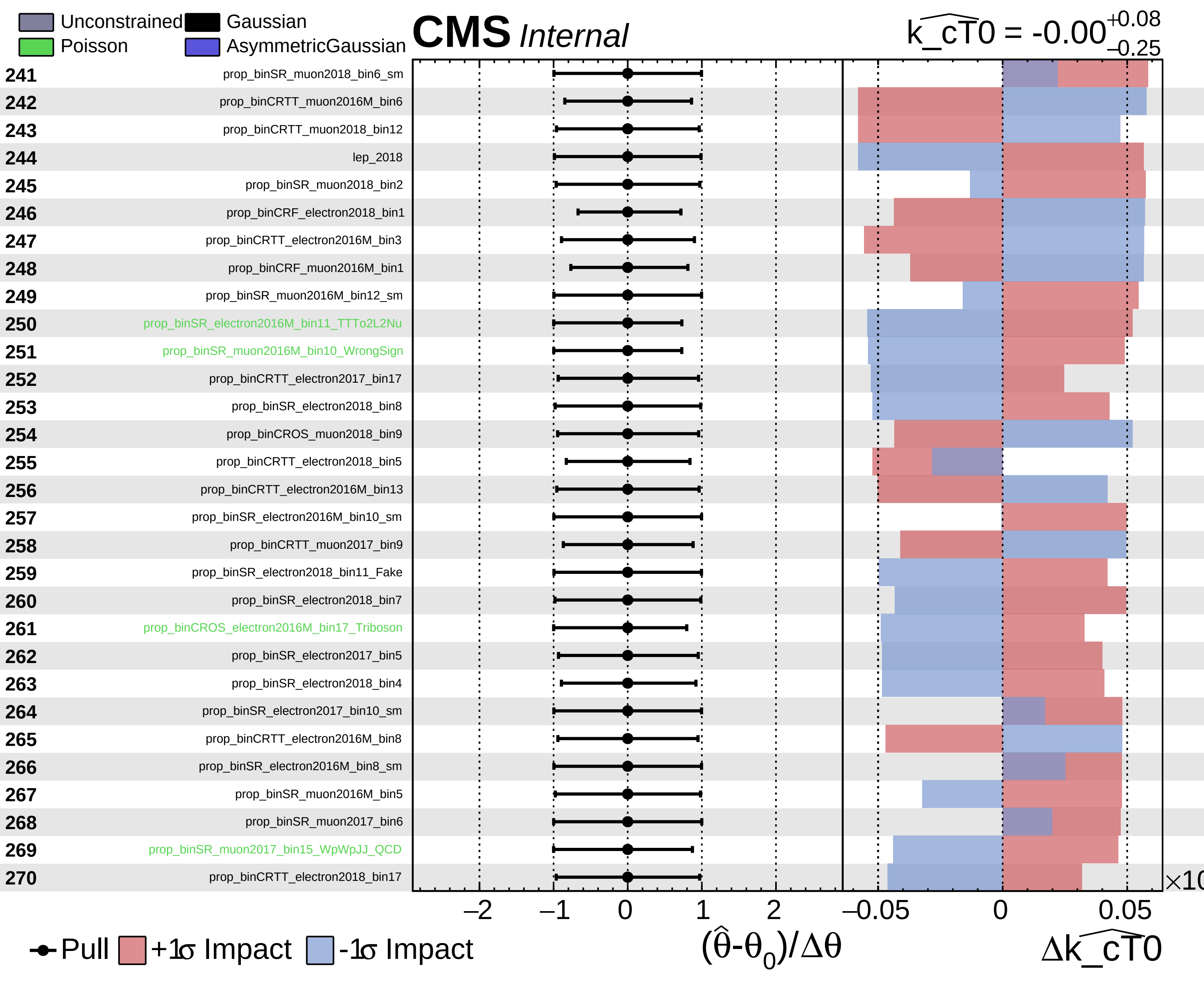
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
151 QCDScale_VG
152 prop_binSR_electron2018_bin10
153 prop_binSR_electron2018_bin12
154 prop_binCROS_electron2018_bin13
155 prop_binCRTT_electron2016M_bin1
156 prop_binCROS_electron2018_bin2
157 prop_binCROS_electron2018_bin1
158 prop_binSR_muon2018_bin11
159 prop_binSR_electron2016M_bin15
160 prop_binSR_muon2017_bin15_TVX
161 prop_binCROS_muon2018_bin0
162 prop_binCROS_electron2018_bin4
163 prop_binSR_muon2018_bin10
164 prop_binCROS_muon2017_bin3
165 prop_binCRF_electron2016M_bin1
166 prop_binCRTT_electron2018_bin12
167 prop_binCRTT_electron2016M_bin0
168 prop_binSR_muon2017_bin15_Triboson
169 prop_binCRTT_electron2016M_bin17
170 prop_binSR_electron2018_bin17_TVX
171 prop_binCROS_electron2018_bin17_VG
172 prop_binCROS_muon2018_bin14_WrongSign
173 prop_binSR_muon2016M_bin14_sm
174 prop_binCROS_electron2017_bin5
175 prop_binSR_muon2018_bin9
176 prop_binCRF_electron2017_bin1
177 prop_binSR_electron2016M_bin14
178 prop_binCROS_electron2017_bin4
179 prop_binCRTT_electron2018_bin16
180 prop_binCROS_electron2017_bin17_TTTo2L2Nu
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



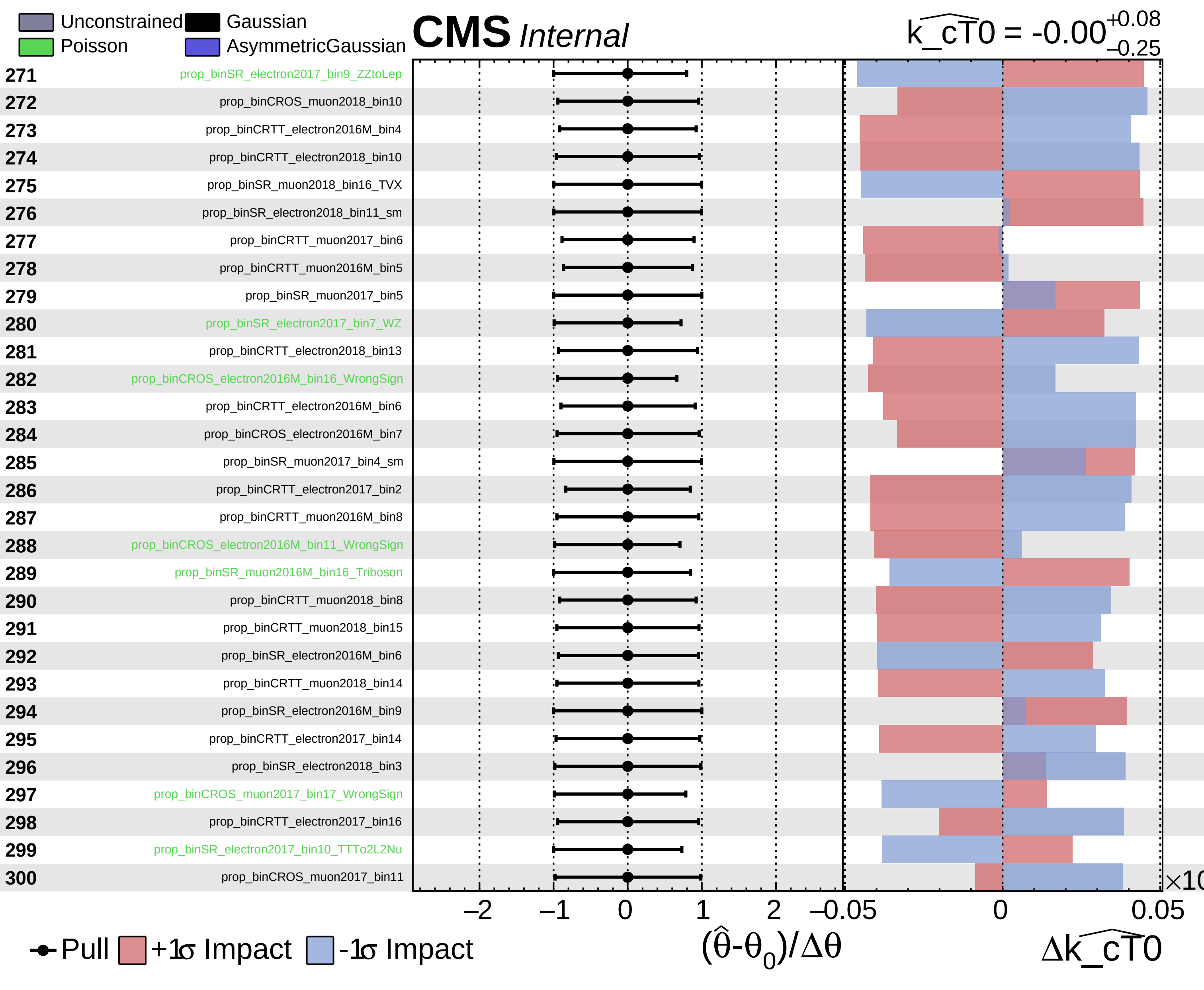
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
181 prop_binCROS_electron2016M_bin8
182 prop_binSR_muon2018_bin3
183 prop_binCRTT_electron2017_bin4
184 prop_binSR_muon2016M_bin10_WZ
185 prop_binCROS_electron2018_bin7_Fake
186 prop_binCROS_muon2017_bin7
187 prop_binCROS_muon2017_bin6
188 prop_binSR_muon2018_bin7_sm
189 prop_binCRF_muon2018_bin7_WrongSign
190 prop_binSR_electron2017_bin9_sm
191 prop_binCROS_muon2018_bin17_TTTo2L2Nu
192 prop_binSR_muon2016M_bin4
193 prop_binCRTT_muon2016M_bin3
194 prop_binCRTT_muon2018_bin17
195 prop_binSR_electron2016M_bin10_TTTo2L2Nu
196 prop_binCRF_muon2018_bin4
197 prop_binCRTT_electron2018_bin11
198 prop_binCRF_muon2018_bin8_sm
199 prop_binCRTT_muon2017_bin11
200 prop_binSR_electron2016M_bin12_sm
201 prop_binCRTT_electron2018_bin8
202 prop_binCROS_muon2018_bin6
203 prop_binSR_electron2016M_bin11_WrongSign
204 prop_binCROS_muon2017_bin2
205 prop_binCROS_electron2018_bin8
206 prop_binSR_muon2016M_bin8_sm
207 prop_binCRF_muon2018_bin9
208 prop_binCROS_electron2017_bin12_WrongSign
209 prop_binSR_electron2017_bin11_sm
210 prop_binSR_electron2018_bin9_sm
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0

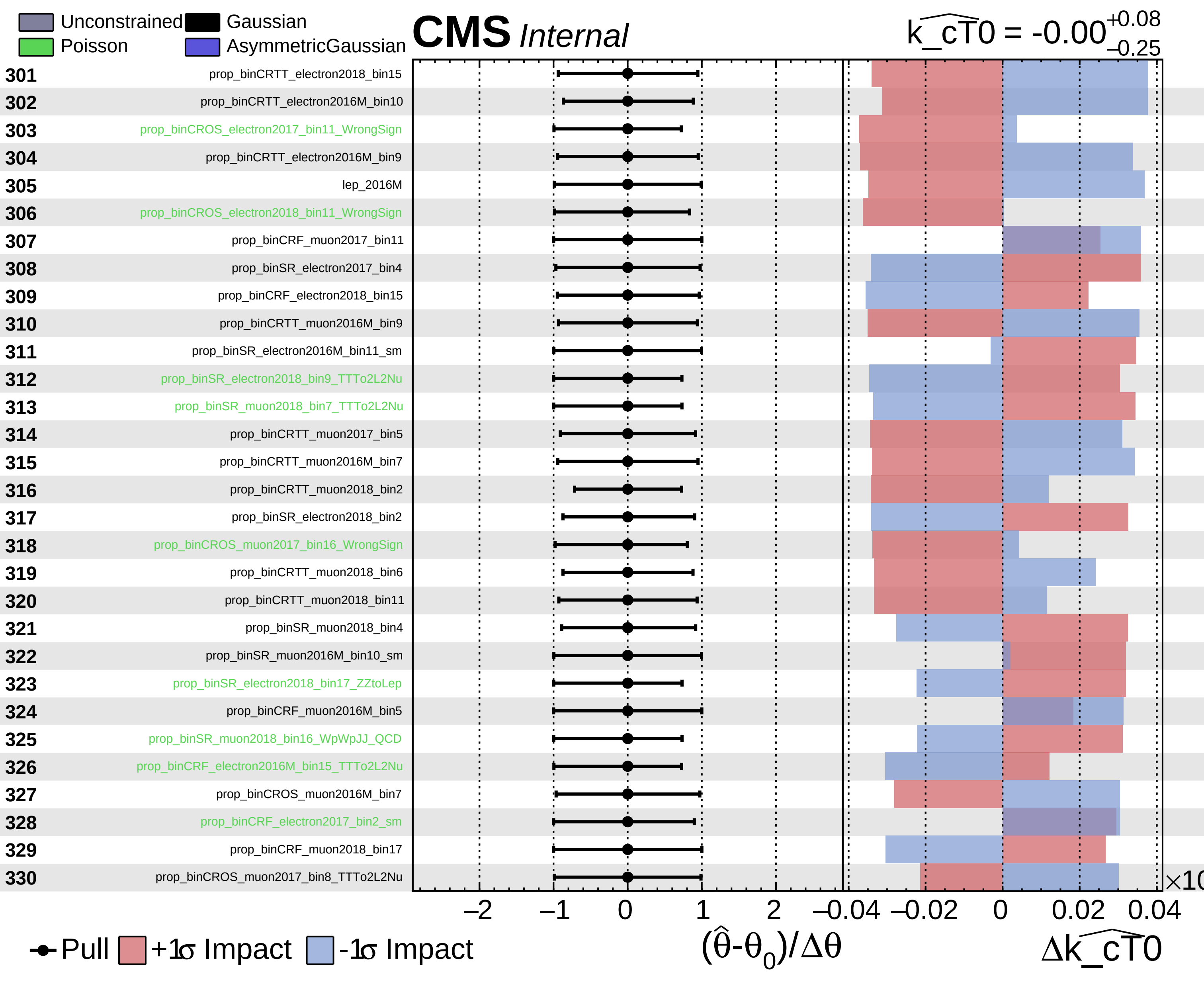


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
211 prop_binCROS_muon2016M_bin17_WrongSign
212 prop_binCROS_electron2016M_bin2
213 prop_binSR_muon2016M_bin11
214 prop_binSR_muon2017_bin11
215 prop_binCRTT_electron2018_bin9
216 prop_binCROS_electron2016M_bin6
217 prop_binCRF_muon2018_bin1
218 prop_binSR_electron2017_bin8
219 prop_binCRTT_muon2017_bin7
220 prop_binCROS_electron2016M_bin15_WrongSign
221 prop_binSR_electron2016M_bin13
222 prop_binCROS_muon2018_bin8
223 prop_binCRTT_muon2018_bin9
224 prop_binSR_electron2018_bin5
225 prop_binCRTT_electron2016M_bin2
226 prop_binCRTT_electron2017_bin5
227 prop_binCRTT_muon2016M_bin12
228 prop_binSR_electron2017_bin13_sm
229 prop_binCRTT_muon2017_bin10
230 prop_binSR_electron2016M_bin11_WZ
231 prop_binSR_muon2017_bin7
232 prop_binCRTT_muon2018_bin5
233 prop_binCROS_muon2016M_bin8
234 prop_binCRTT_electron2016M_bin7
235 prop_binSR_electron2017_bin12
236 prop_binCRTT_muon2016M_bin4
237 prop_binSR_muon2017_bin9
238 prop_binSR_muon2017_bin10_sm
239 prop_binCRTT_muon2017_bin4
240 prop_binCROS_muon2018_bin11
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT0

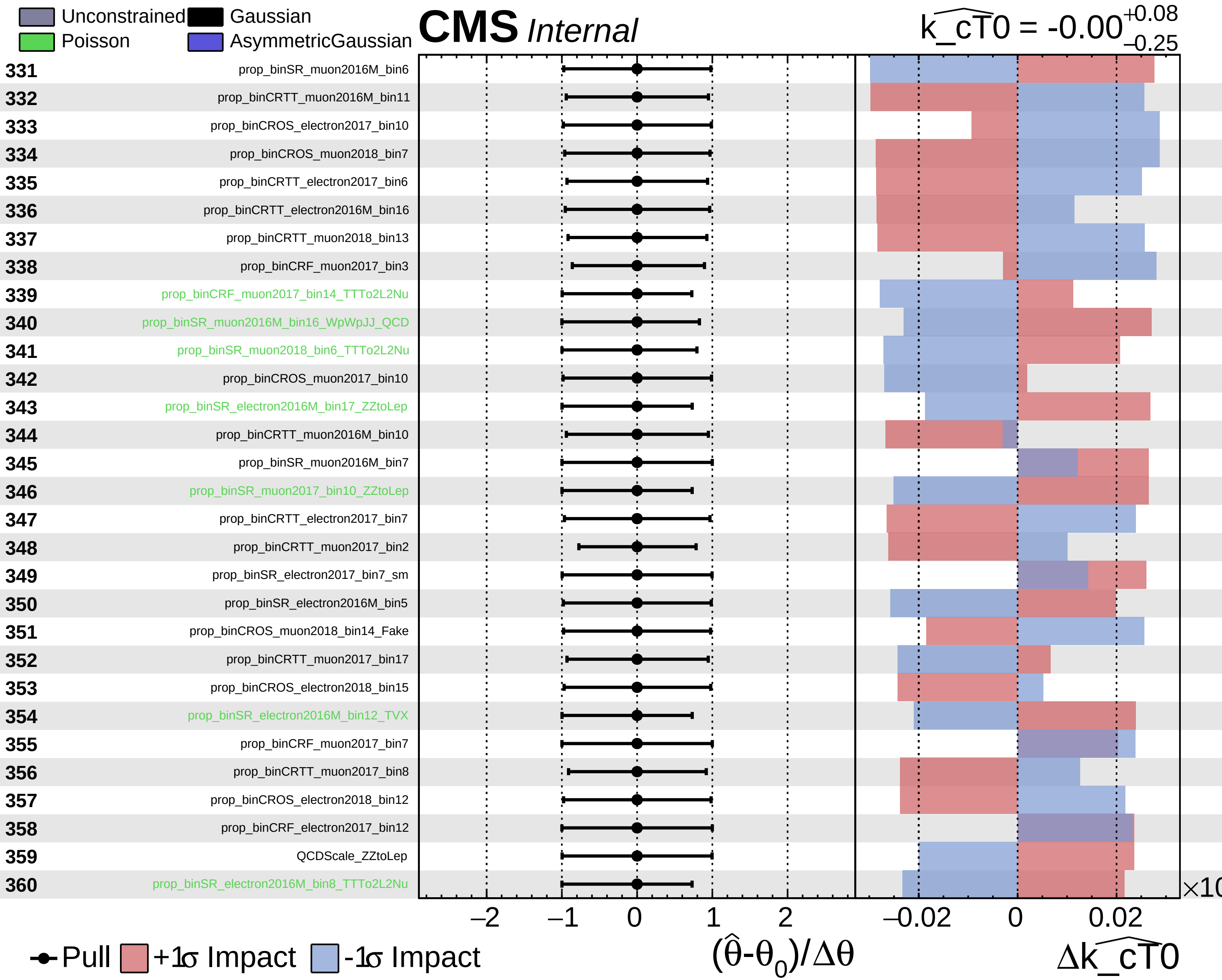


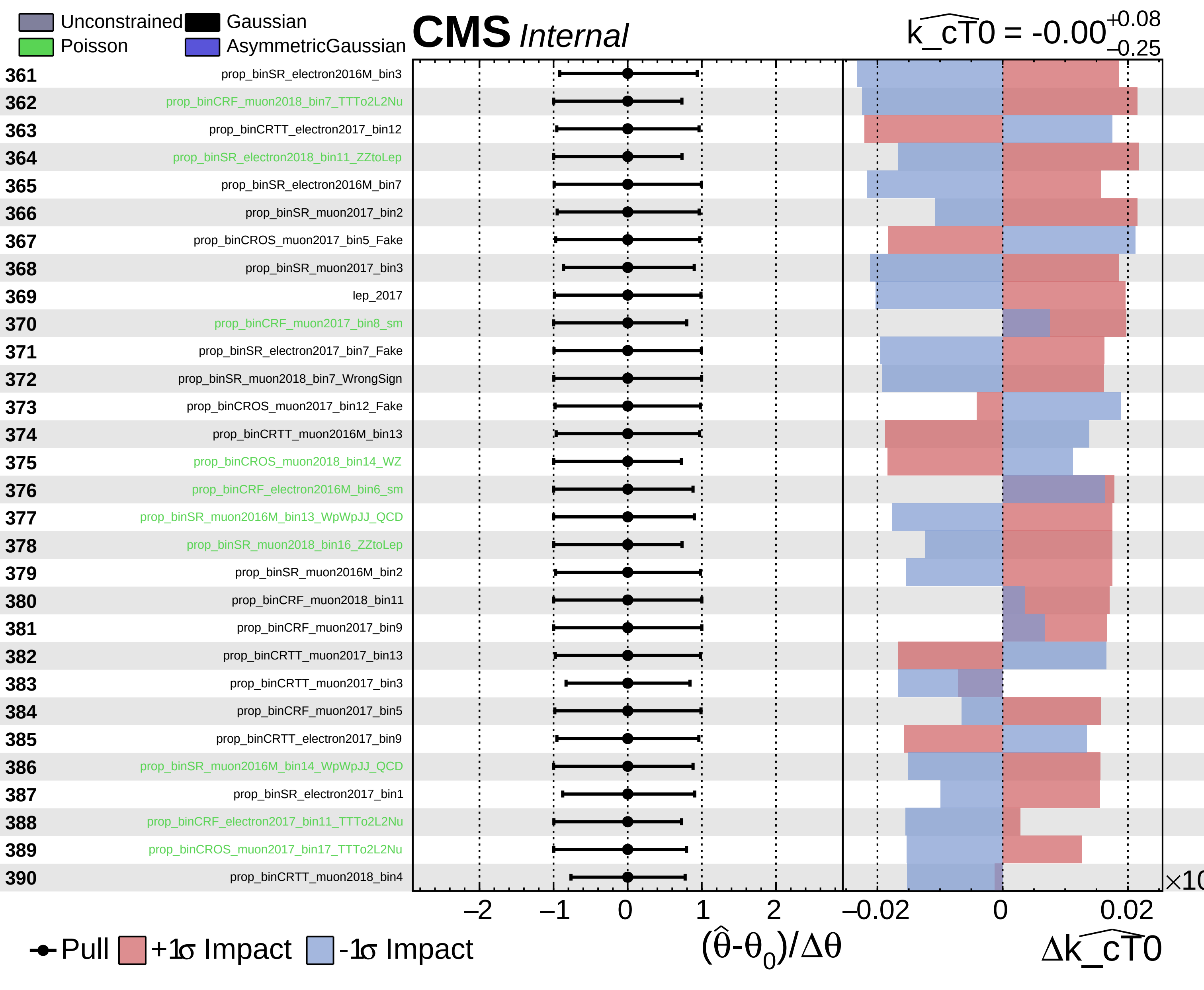
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
241 prop_binSR_muon2018_bin6_sm
242 prop_binCRTT_muon2016M_bin6
243 prop_binCRTT_muon2018_bin12
244 lep_2018
245 prop_binSR_muon2018_bin2
246 prop_binCRF_electron2018_bin1
247 prop_binCRTT_electron2016M_bin3
248 prop_binCRF_muon2016M_bin1
249 prop_binSR_muon2016M_bin12_sm
250 prop_binSR_electron2016M_bin11_TTTo2L2Nu
251 prop_binSR_muon2016M_bin10_WrongSign
252 prop_binCRTT_electron2017_bin17
253 prop_binSR_electron2018_bin8
254 prop_binCROS_muon2018_bin9
255 prop_binCRTT_electron2018_bin5
256 prop_binCRTT_electron2016M_bin13
257 prop_binSR_electron2016M_bin10_sm
258 prop_binCRTT_muon2017_bin9
259 prop_binSR_electron2018_bin11_Fake
260 prop_binSR_electron2018_bin7
261 prop_binCROS_electron2016M_bin17_Triboson
262 prop_binSR_electron2017_bin5
263 prop_binSR_electron2018_bin4
264 prop_binSR_electron2017_bin10_sm
265 prop_binCRTT_electron2016M_bin8
266 prop_binSR_electron2016M_bin8_sm
267 prop_binSR_muon2016M_bin5
268 prop_binSR_muon2017_bin6
269 prop_binSR_muon2017_bin15_WpWpJJ_QCD
270 prop_binCRTT_electron2018_bin17
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
271 prop_binSR_electron2017_bin9_ZZtoLep
272 prop_binCROS_muon2018_bin10
273 prop_binCRTT_electron2016M_bin4
274 prop_binCRTT_electron2018_bin10
275 prop_binSR_muon2018_bin16_TVX
276 prop_binSR_electron2018_bin11_sm
277 prop_binCRTT_muon2017_bin6
278 prop_binCRTT_muon2016M_bin5
279 prop_binSR_muon2017_bin5
280 prop_binSR_electron2017_bin7_WZ
281 prop_binCRTT_electron2018_bin13
282 prop_binCROS_electron2016M_bin16_WrongSign
283 prop_binCRTT_electron2016M_bin6
284 prop_binCROS_electron2016M_bin7
285 prop_binSR_muon2017_bin4_sm
286 prop_binCRTT_electron2017_bin2
287 prop_binCRTT_muon2016M_bin8
288 prop_binCROS_electron2016M_bin11_WrongSign
289 prop_binSR_muon2016M_bin16_Triboson
290 prop_binCRTT_muon2018_bin8
291 prop_binCRTT_muon2018_bin15
292 prop_binSR_electron2016M_bin6
293 prop_binCRTT_muon2018_bin14
294 prop_binSR_electron2016M_bin9
295 prop_binCRTT_electron2017_bin14
296 prop_binSR_electron2018_bin3
297 prop_binCROS_muon2017_bin17_WrongSign
298 prop_binCRTT_electron2017_bin16
299 prop_binSR_electron2017_bin10_TTTo2L2Nu
300 prop_binCROS_muon2017_bin11
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0

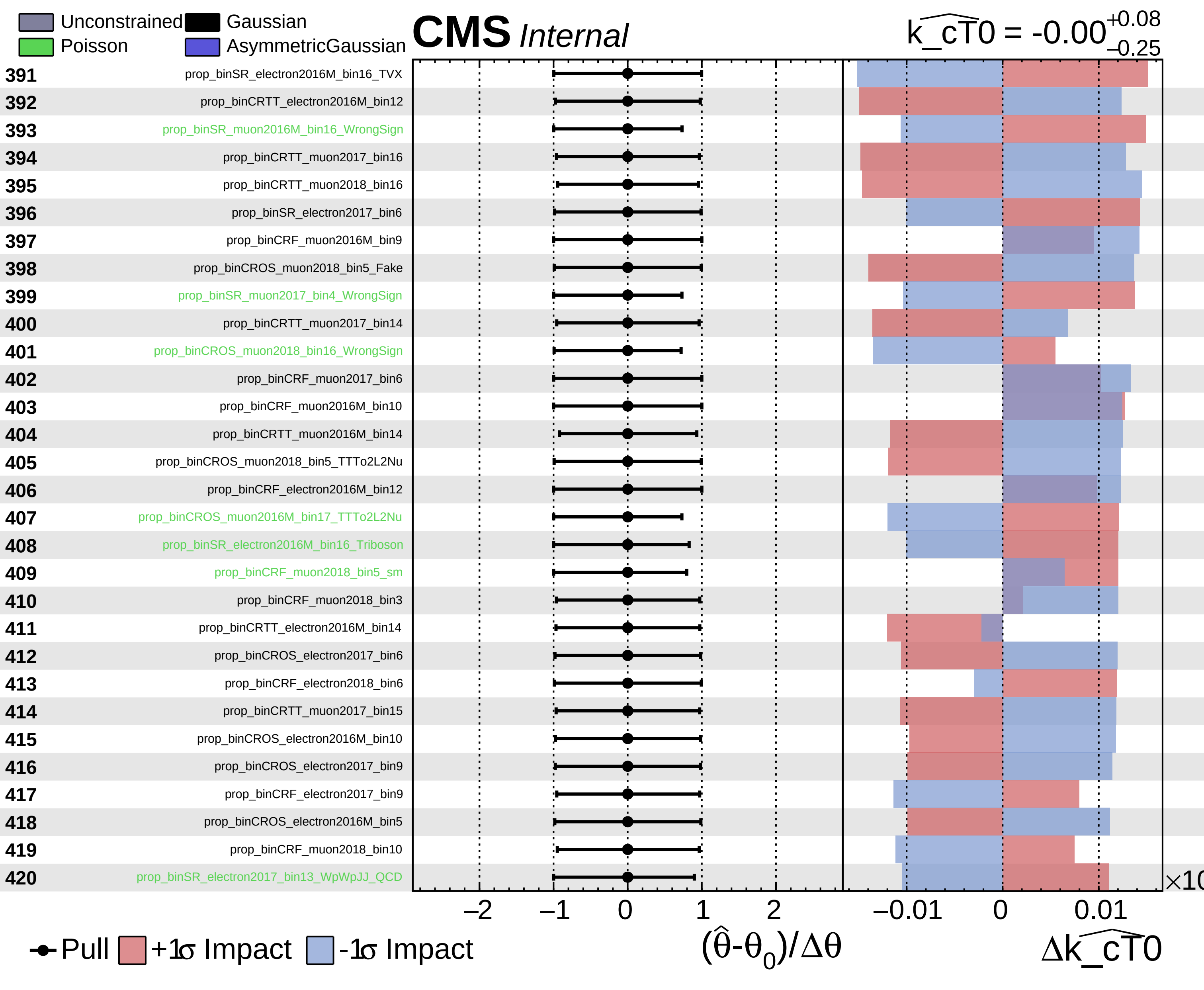


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
301 prop_binCRTT_electron2018_bin15
302 prop_binCRTT_electron2016M_bin10
303 prop_binCROS_electron2017_bin11_WrongSign
304 prop_binCRTT_electron2016M_bin9
305 lep_2016M
306 prop_binCROS_electron2018_bin11_WrongSign
307 prop_binCRF_muon2017_bin11
308 prop_binSR_electron2017_bin4
309 prop_binCRF_electron2018_bin15
310 prop_binCRTT_muon2016M_bin9
311 prop_binSR_electron2016M_bin11_sm
312 prop_binSR_electron2018_bin9_TTTo2L2Nu
313 prop_binSR_muon2018_bin7_TTTo2L2Nu
314 prop_binCRTT_muon2017_bin5
315 prop_binCRTT_muon2016M_bin7
316 prop_binCRTT_muon2018_bin2
317 prop_binSR_electron2018_bin2
318 prop_binCROS_muon2017_bin16_WrongSign
319 prop_binCRTT_muon2018_bin6
320 prop_binCRTT_muon2018_bin11
321 prop_binSR_muon2018_bin4
322 prop_binSR_muon2016M_bin10_sm
323 prop_binSR_electron2018_bin17_ZZtoLep
324 prop_binCRF_muon2016M_bin5
325 prop_binSR_muon2018_bin16_WpWpJJ_QCD
326 prop_binCRF_electron2016M_bin15_TTTo2L2Nu
327 prop_binCROS_muon2016M_bin7
328 prop_binCRF_electron2017_bin2_sm
329 prop_binCRF_muon2018_bin17
330 prop_binCROS_muon2017_bin8_TTTo2L2Nu
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0

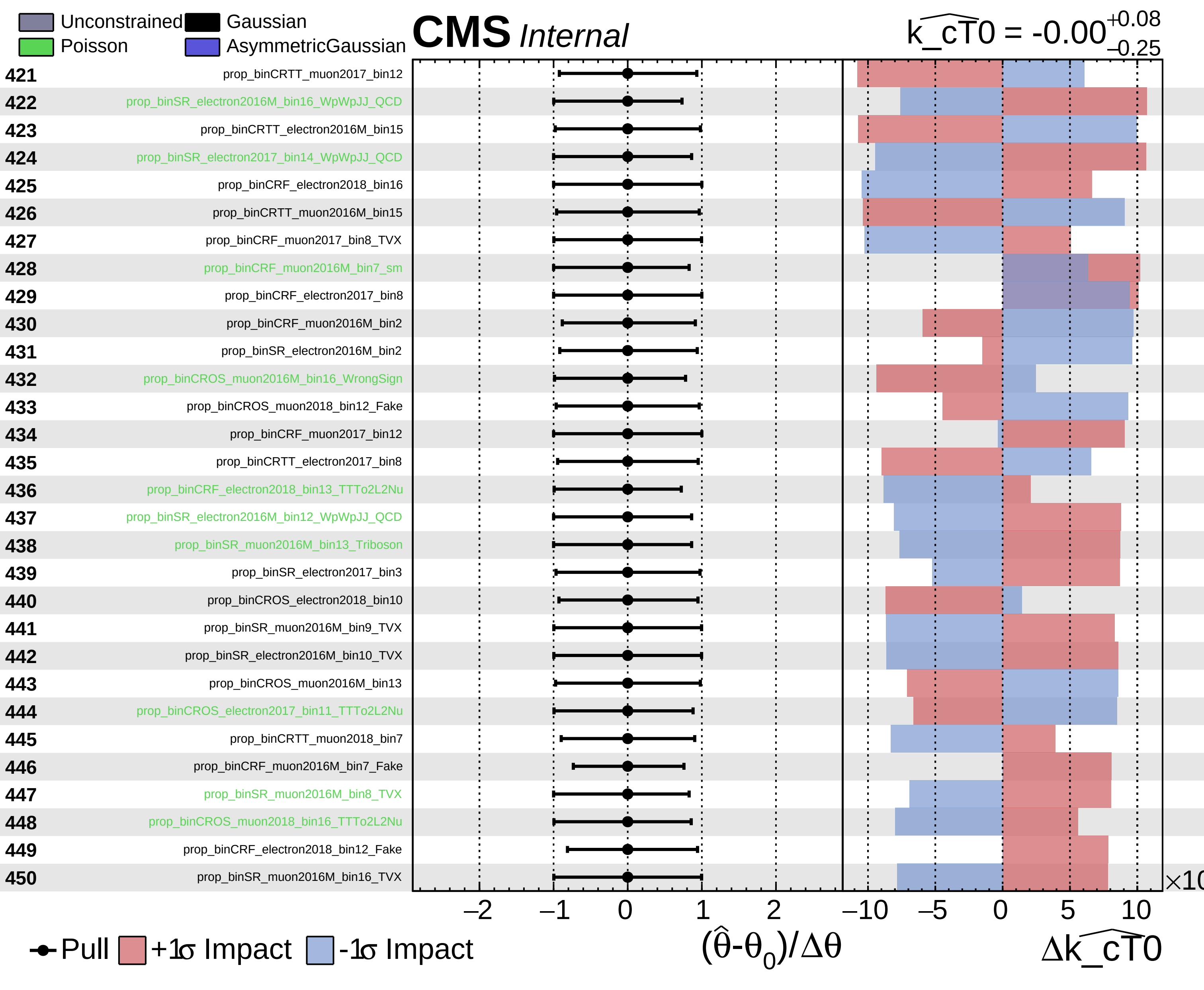

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
331 prop_binSR_muon2016M_bin6
332 prop_binCRTT_muon2016M_bin11
333 prop_binCROS_electron2017_bin10
334 prop_binCROS_muon2018_bin7
335 prop_binCRTT_electron2017_bin6
336 prop_binCRTT_electron2016M_bin16
337 prop_binCRTT_muon2018_bin13
338 prop_binCRF_muon2017_bin3
339 prop_binCRF_muon2017_bin14_TTTo2L2Nu
340 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
341 prop_binSR_muon2018_bin6_TTTo2L2Nu
342 prop_binCROS_muon2017_bin10
343 prop_binSR_electron2016M_bin17_ZZtoLep
344 prop_binCRTT_muon2016M_bin10
345 prop_binSR_muon2016M_bin7
346 prop_binSR_muon2017_bin10_ZZtoLep
347 prop_binCRTT_electron2017_bin7
348 prop_binCRTT_muon2017_bin2
349 prop_binSR_electron2017_bin7_sm
350 prop_binSR_electron2016M_bin5
351 prop_binCROS_muon2018_bin14_Fake
352 prop_binCRTT_muon2017_bin17
353 prop_binCROS_electron2018_bin15
354 prop_binSR_electron2016M_bin12_TVX
355 prop_binCRF_muon2017_bin7
356 prop_binCRTT_muon2017_bin8
357 prop_binCROS_electron2018_bin12
358 prop_binCRF_electron2017_bin12
359 QCDScale_ZZtoLep
360 prop_binSR_electron2016M_bin8_TTTo2L2Nu
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0



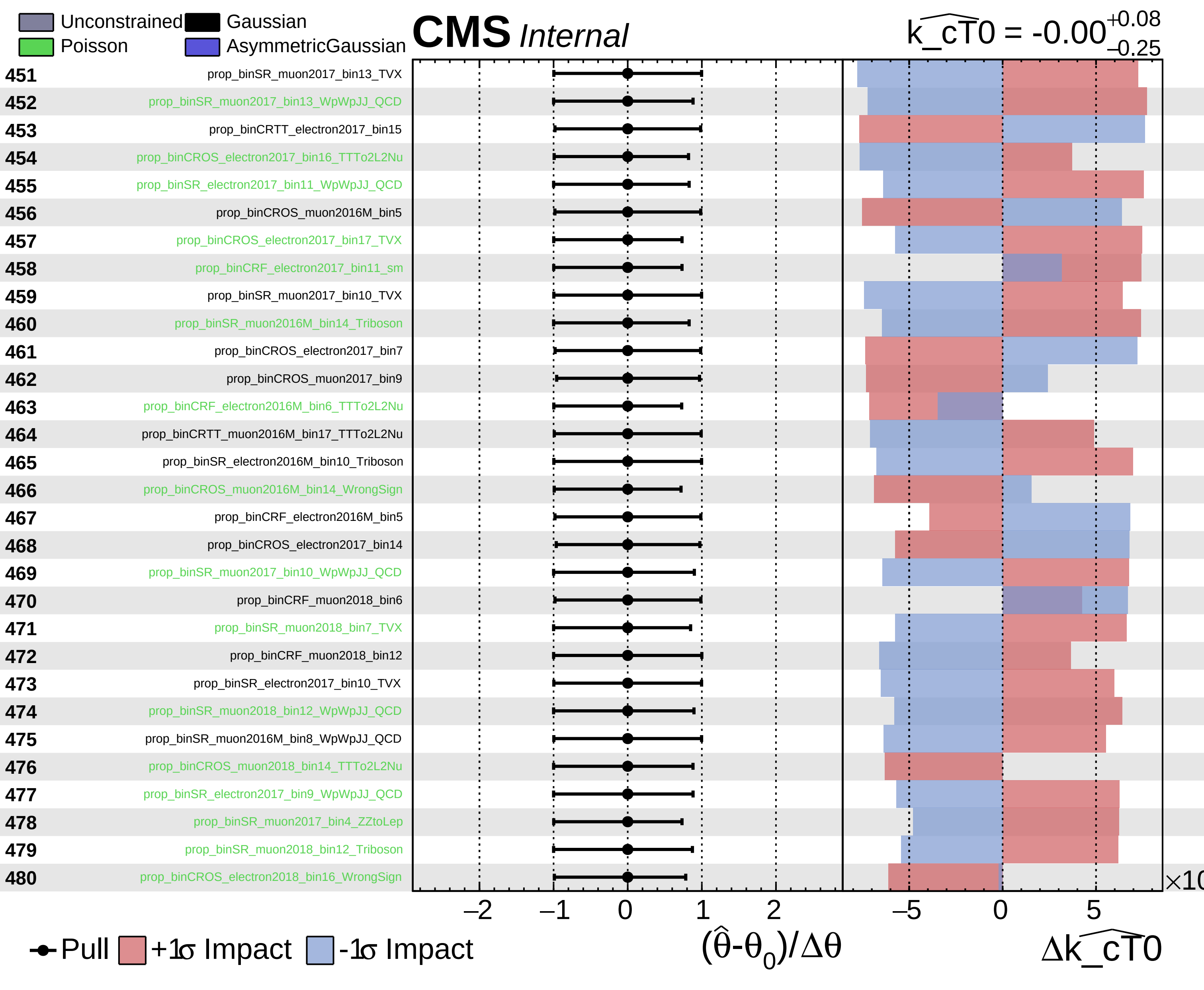
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
361 prop_binSR_electron2016M_bin3
362 prop_binCRF_muon2018_bin7_TTTo2L2Nu
363 prop_binCRTT_electron2017_bin12
364 prop_binSR_electron2018_bin11_ZZtoLep
365 prop_binSR_electron2016M_bin7
366 prop_binSR_muon2017_bin2
367 prop_binCROS_muon2017_bin5_Fake
368 prop_binSR_muon2017_bin3
369 lep_2017
370 prop_binCRF_muon2017_bin8_sm
371 prop_binSR_electron2017_bin7_Fake
372 prop_binSR_muon2018_bin7_WrongSign
373 prop_binCROS_muon2017_bin12_Fake
374 prop_binCRTT_muon2016M_bin13
375 prop_binCROS_muon2018_bin14_WZ
376 prop_binCRF_electron2016M_bin6_sm
377 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
378 prop_binSR_muon2018_bin16_ZZtoLep
379 prop_binSR_muon2016M_bin2
380 prop_binCRF_muon2018_bin11
381 prop_binCRF_muon2017_bin9
382 prop_binCRTT_muon2017_bin13
383 prop_binCRTT_muon2017_bin3
384 prop_binCRF_muon2017_bin5
385 prop_binCRTT_electron2017_bin9
386 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
387 prop_binSR_electron2017_bin1
388 prop_binCRF_electron2017_bin11_TTTo2L2Nu
389 prop_binCROS_muon2017_bin17_TTTo2L2Nu
390 prop_binCRTT_muon2018_bin4
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT0



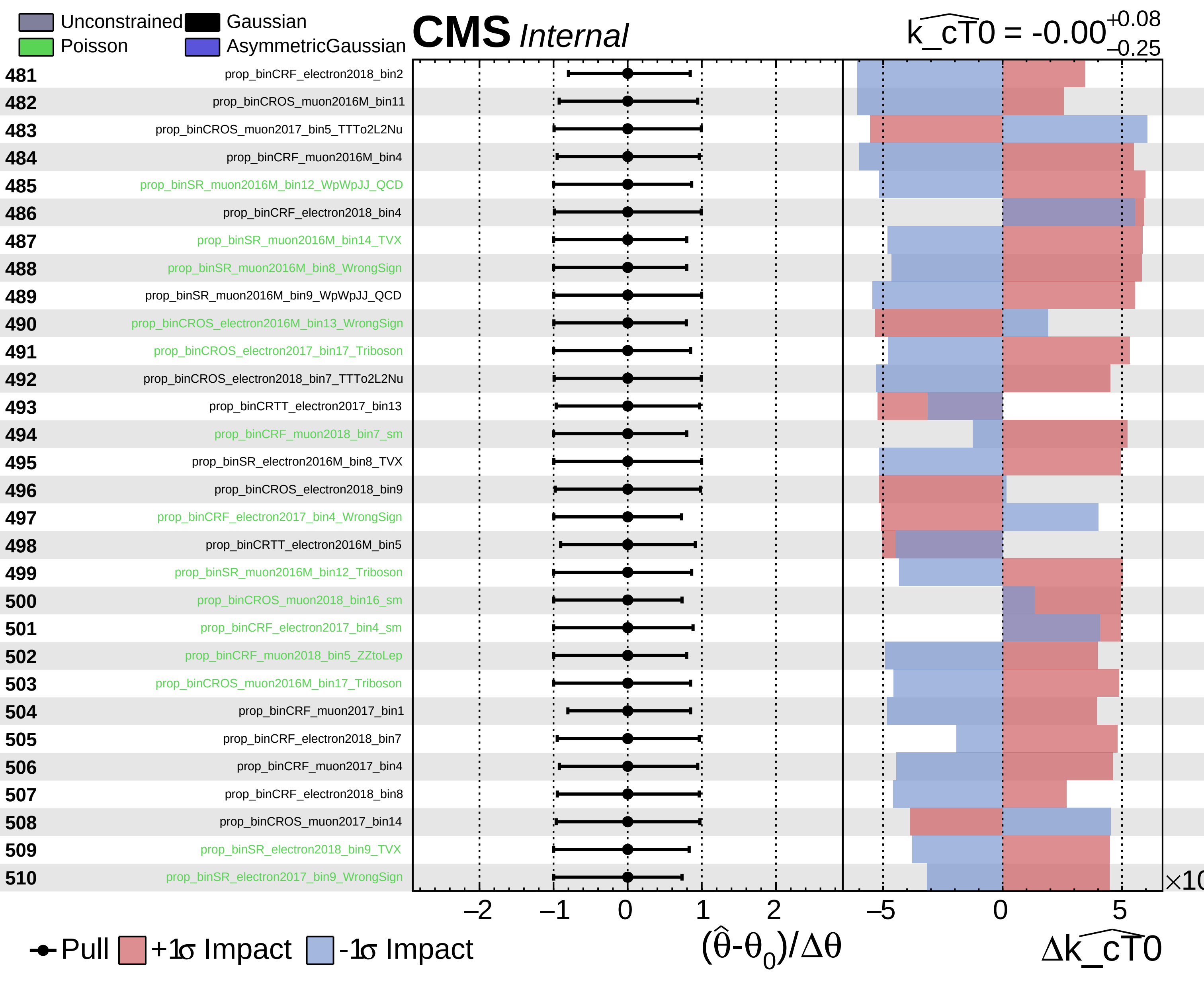
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
391 prop_binSR_electron2016M_bin16_TVX
392 prop_binCRTT_electron2016M_bin12
393 prop_binSR_muon2016M_bin16_WrongSign
394 prop_binCRTT_muon2017_bin16
395 prop_binCRTT_muon2018_bin16
396 prop_binSR_electron2017_bin6
397 prop_binCRF_muon2016M_bin9
398 prop_binCROS_muon2018_bin5_Fake
399 prop_binSR_muon2017_bin4_WrongSign
400 prop_binCRTT_muon2017_bin14
401 prop_binCROS_muon2018_bin16_WrongSign
402 prop_binCRF_muon2017_bin6
403 prop_binCRF_muon2016M_bin10
404 prop_binCRTT_muon2016M_bin14
405 prop_binCROS_muon2018_bin5_TTTo2L2Nu
406 prop_binCRF_electron2016M_bin12
407 prop_binCROS_muon2016M_bin17_TTTo2L2Nu
408 prop_binSR_electron2016M_bin16_Triboson
409 prop_binCRF_muon2018_bin5_sm
410 prop_binCRF_muon2018_bin3
411 prop_binCRTT_electron2016M_bin14
412 prop_binCROS_electron2017_bin6
413 prop_binCRF_electron2018_bin6
414 prop_binCRTT_muon2017_bin15
415 prop_binCROS_electron2016M_bin10
416 prop_binCROS_electron2017_bin9
417 prop_binCRF_electron2017_bin9
418 prop_binCROS_electron2016M_bin5
419 prop_binCRF_muon2018_bin10
420 prop_binSR_electron2017_bin13_WpWpJJ_QCD
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



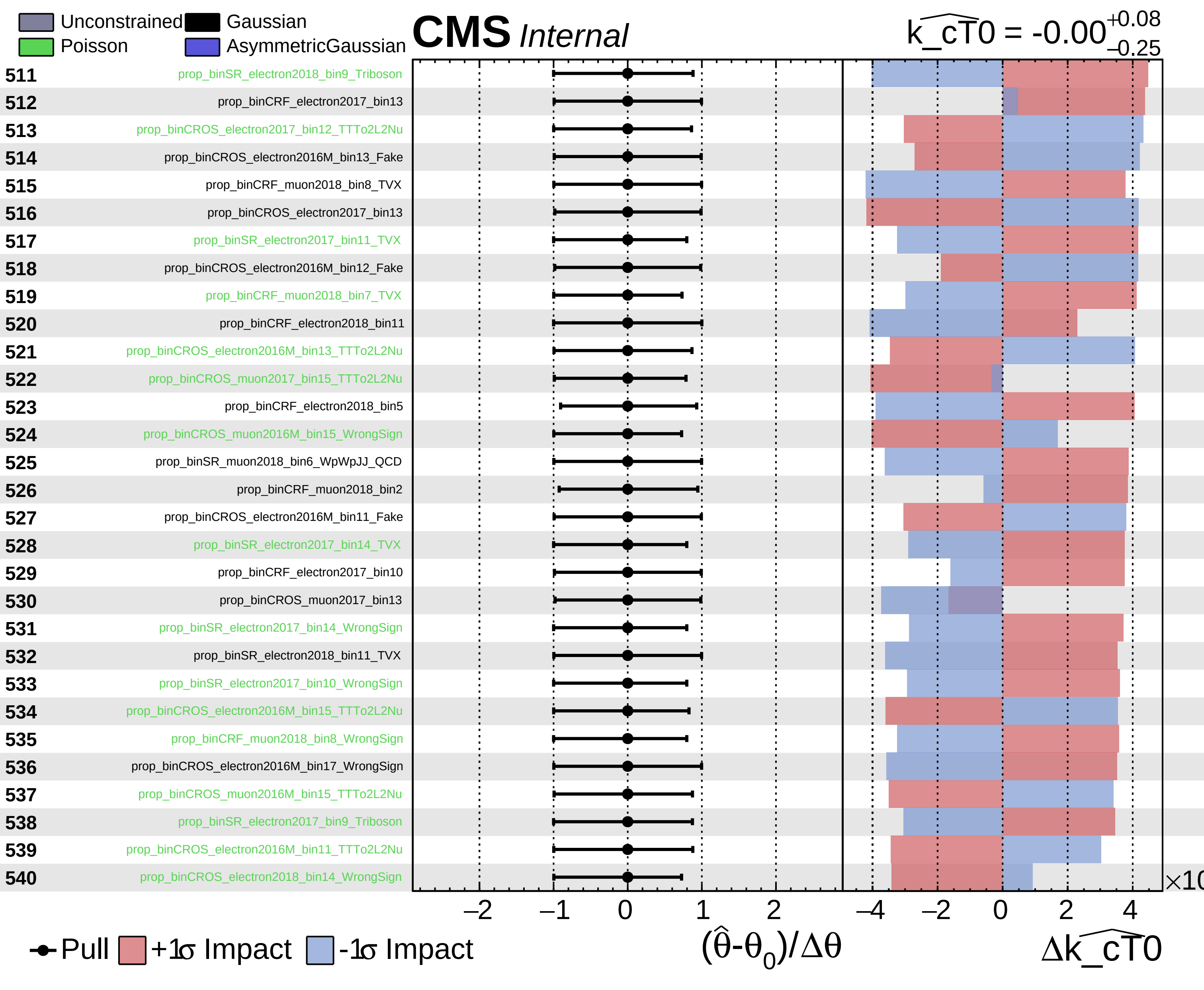
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
421 prop_binCRTT_muon2017_bin12
422 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
423 prop_binCRTT_electron2016M_bin15
424 prop_binSR_electron2017_bin14_WpWpJJ_QCD
425 prop_binCRF_electron2018_bin16
426 prop_binCRTT_muon2016M_bin15
427 prop_binCRF_muon2017_bin8_TVX
428 prop_binCRF_muon2016M_bin7_sm
429 prop_binCRF_electron2017_bin8
430 prop_binCRF_muon2016M_bin2
431 prop_binSR_electron2016M_bin2
432 prop_binCROS_muon2016M_bin16_WrongSign
433 prop_binCROS_muon2018_bin12_Fake
434 prop_binCRF_muon2017_bin12
435 prop_binCRTT_electron2017_bin8
436 prop_binCRF_electron2018_bin13_TTTo2L2Nu
437 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
438 prop_binSR_muon2016M_bin13_Triboson
439 prop_binSR_electron2017_bin3
440 prop_binCROS_electron2018_bin10
441 prop_binSR_muon2016M_bin9_TVX
442 prop_binSR_electron2016M_bin10_TVX
443 prop_binCROS_muon2016M_bin13
444 prop_binCROS_electron2017_bin11_TTTo2L2Nu
445 prop_binCRTT_muon2018_bin7
446 prop_binCRF_muon2016M_bin7_Fake
447 prop_binSR_muon2016M_bin8_TVX
448 prop_binCROS_muon2018_bin16_TTTo2L2Nu
449 prop_binCRF_electron2018_bin12_Fake
450 prop_binSR_muon2016M_bin16_TVX
−2 −1 0 1 2
−10 −5 0 5 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

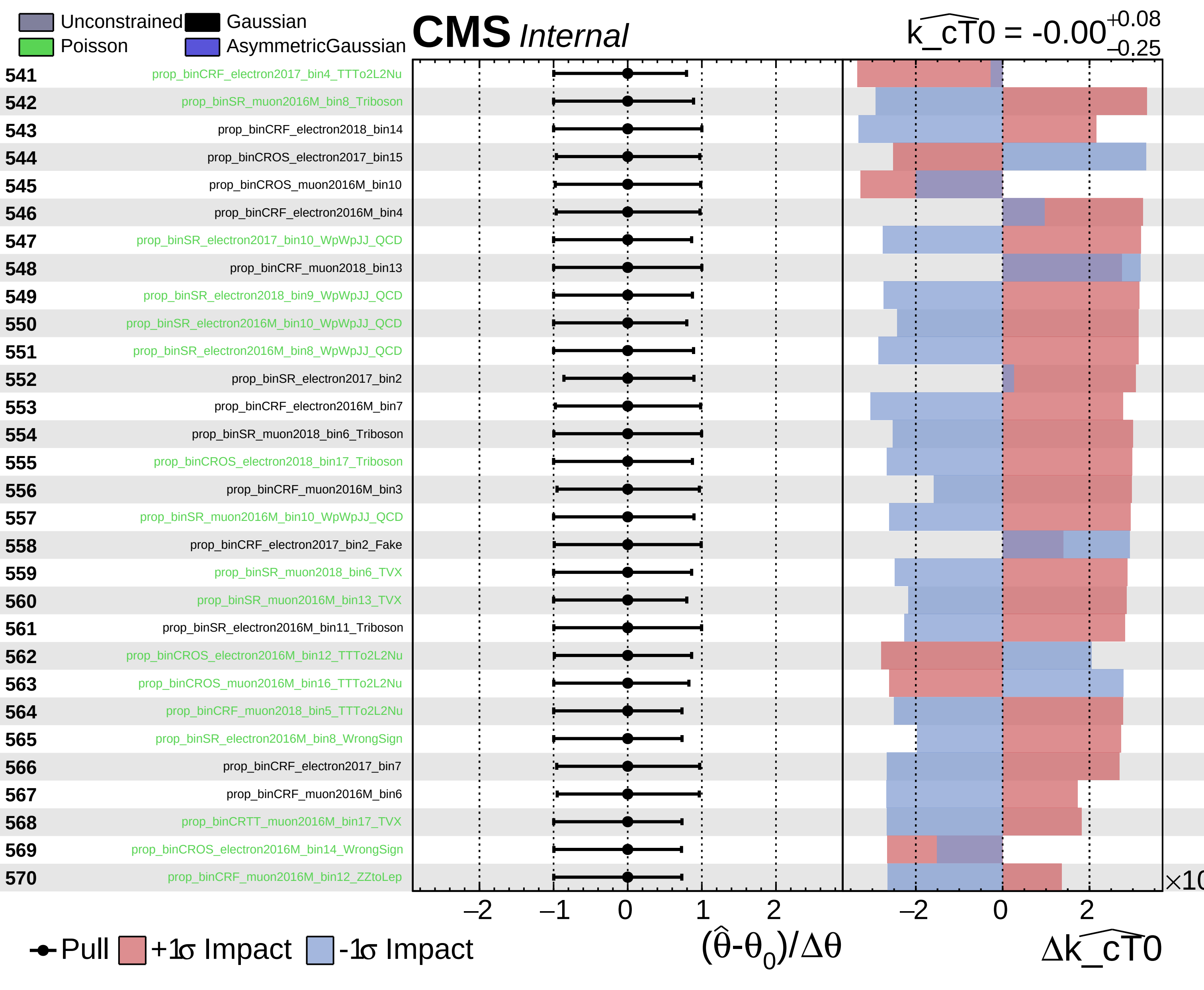


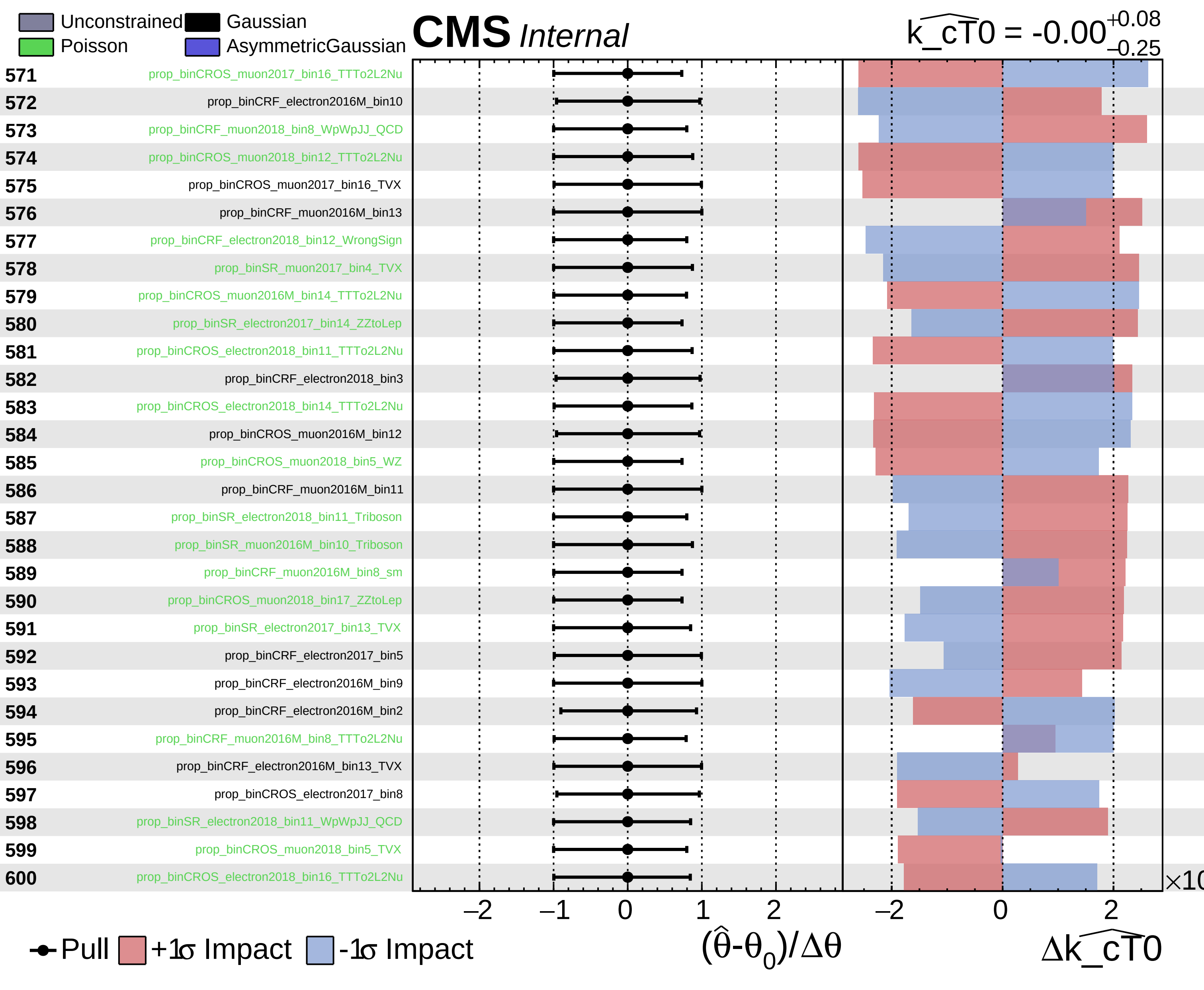
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
451 prop_binSR_muon2017_bin13_TVX
452 prop_binSR_muon2017_bin13_WpWpJJ_QCD
453 prop_binCRTT_electron2017_bin15
454 prop_binCROS_electron2017_bin16_TTTo2L2Nu
455 prop_binSR_electron2017_bin11_WpWpJJ_QCD
456 prop_binCROS_muon2016M_bin5
457 prop_binCROS_electron2017_bin17_TVX
458 prop_binCRF_electron2017_bin11_sm
459 prop_binSR_muon2017_bin10_TVX
460 prop_binSR_muon2016M_bin14_Triboson
461 prop_binCROS_electron2017_bin7
462 prop_binCROS_muon2017_bin9
463 prop_binCRF_electron2016M_bin6_TTTo2L2Nu
464 prop_binCRTT_muon2016M_bin17_TTTo2L2Nu
465 prop_binSR_electron2016M_bin10_Triboson
466 prop_binCROS_muon2016M_bin14_WrongSign
467 prop_binCRF_electron2016M_bin5
468 prop_binCROS_electron2017_bin14
469 prop_binSR_muon2017_bin10_WpWpJJ_QCD
470 prop_binCRF_muon2018_bin6
471 prop_binSR_muon2018_bin7_TVX
472 prop_binCRF_muon2018_bin12
473 prop_binSR_electron2017_bin10_TVX
474 prop_binSR_muon2018_bin12_WpWpJJ_QCD
475 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
476 prop_binCROS_muon2018_bin14_TTTo2L2Nu
477 prop_binSR_electron2017_bin9_WpWpJJ_QCD
478 prop_binSR_muon2017_bin4_ZZtoLep
479 prop_binSR_muon2018_bin12_Triboson
480 prop_binCROS_electron2018_bin16_WrongSign
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

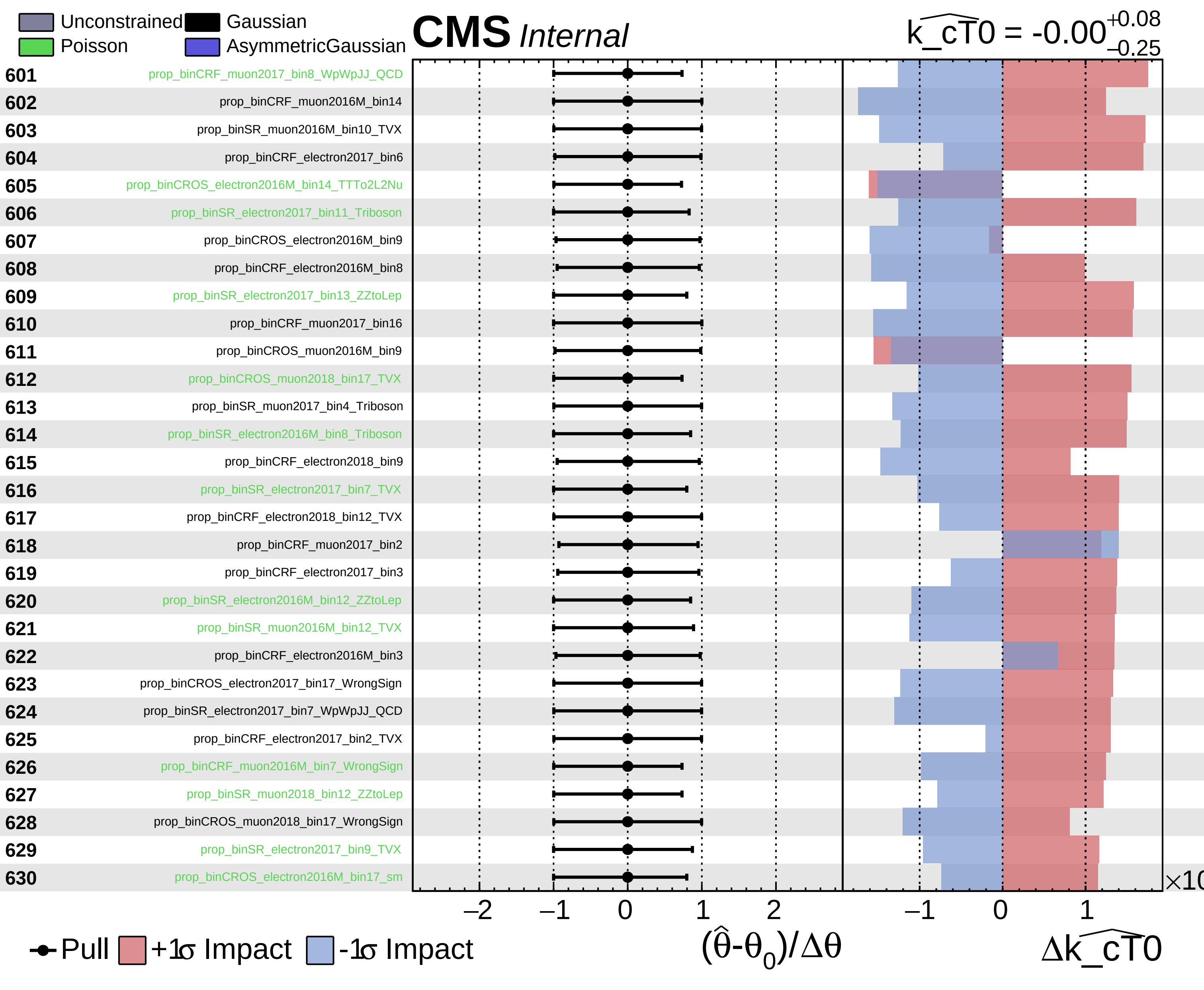


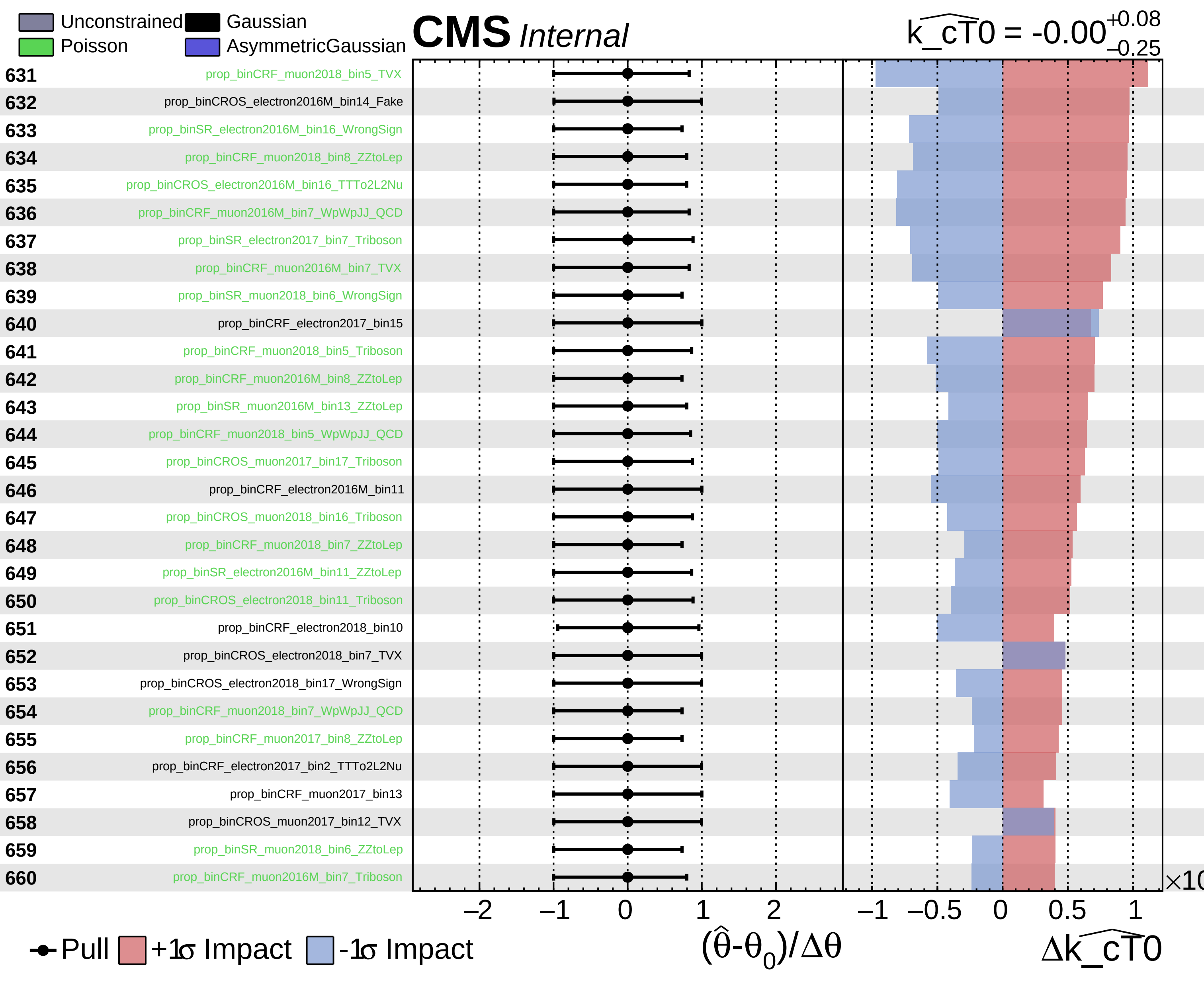
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
481 prop_binCRF_electron2018_bin2
482 prop_binCROS_muon2016M_bin11
483 prop_binCROS_muon2017_bin5_TTTo2L2Nu
484 prop_binCRF_muon2016M_bin4
485 prop_binSR_muon2016M_bin12_WpWpJJ_QCD
486 prop_binCRF_electron2018_bin4
487 prop_binSR_muon2016M_bin14_TVX
488 prop_binSR_muon2016M_bin8_WrongSign
489 prop_binSR_muon2016M_bin9_WpWpJJ_QCD
490 prop_binCROS_electron2016M_bin13_WrongSign
491 prop_binCROS_electron2017_bin17_Triboson
492 prop_binCROS_electron2018_bin7_TTTo2L2Nu
493 prop_binCRTT_electron2017_bin13
494 prop_binCRF_muon2018_bin7_sm
495 prop_binSR_electron2016M_bin8_TVX
496 prop_binCROS_electron2018_bin9
497 prop_binCRF_electron2017_bin4_WrongSign
498 prop_binCRTT_electron2016M_bin5
499 prop_binSR_muon2016M_bin12_Triboson
500 prop_binCROS_muon2018_bin16_sm
501 prop_binCRF_electron2017_bin4_sm
502 prop_binCRF_muon2018_bin5_ZZtoLep
503 prop_binCROS_muon2016M_bin17_Triboson
504 prop_binCRF_muon2017_bin1
505 prop_binCRF_electron2018_bin7
506 prop_binCRF_muon2017_bin4
507 prop_binCRF_electron2018_bin8
508 prop_binCROS_muon2017_bin14
509 prop_binSR_electron2018_bin9_TVX
510 prop_binSR_electron2017_bin9_WrongSign
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
511 prop_binSR_electron2018_bin9_Triboson
512 prop_binCRF_electron2017_bin13
513 prop_binCROS_electron2017_bin12_TTTo2L2Nu
514 prop_binCROS_electron2016M_bin13_Fake
515 prop_binCRF_muon2018_bin8_TVX
516 prop_binCROS_electron2017_bin13
517 prop_binSR_electron2017_bin11_TVX
518 prop_binCROS_electron2016M_bin12_Fake
519 prop_binCRF_muon2018_bin7_TVX
520 prop_binCRF_electron2018_bin11
521 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
522 prop_binCROS_muon2017_bin15_TTTo2L2Nu
523 prop_binCRF_electron2018_bin5
524 prop_binCROS_muon2016M_bin15_WrongSign
525 prop_binSR_muon2018_bin6_WpWpJJ_QCD
526 prop_binCRF_muon2018_bin2
527 prop_binCROS_electron2016M_bin11_Fake
528 prop_binSR_electron2017_bin14_TVX
529 prop_binCRF_electron2017_bin10
530 prop_binCROS_muon2017_bin13
531 prop_binSR_electron2017_bin14_WrongSign
532 prop_binSR_electron2018_bin11_TVX
533 prop_binSR_electron2017_bin10_WrongSign
534 prop_binCROS_electron2016M_bin15_TTTo2L2Nu
535 prop_binCRF_muon2018_bin8_WrongSign
536 prop_binCROS_electron2016M_bin17_WrongSign
537 prop_binCROS_muon2016M_bin15_TTTo2L2Nu
538 prop_binSR_electron2017_bin9_Triboson
539 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
540 prop_binCROS_electron2018_bin14_WrongSign
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

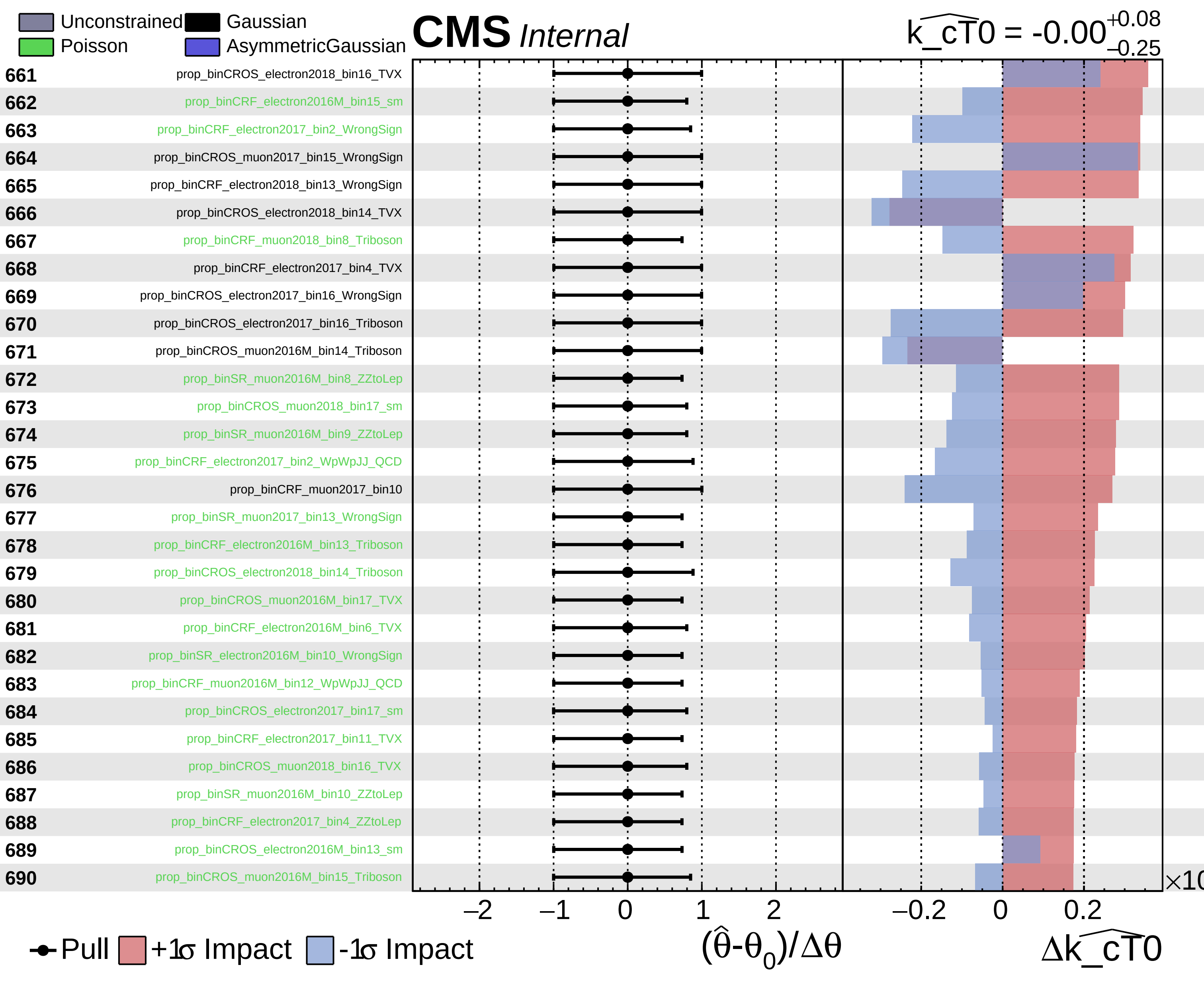

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
541 prop_binCRF_electron2017_bin4_TTTo2L2Nu
542 prop_binSR_muon2016M_bin8_Triboson
543 prop_binCRF_electron2018_bin14
544 prop_binCROS_electron2017_bin15
545 prop_binCROS_muon2016M_bin10
546 prop_binCRF_electron2016M_bin4
547 prop_binSR_electron2017_bin10_WpWpJJ_QCD
548 prop_binCRF_muon2018_bin13
549 prop_binSR_electron2018_bin9_WpWpJJ_QCD
550 prop_binSR_electron2016M_bin10_WpWpJJ_QCD
551 prop_binSR_electron2016M_bin8_WpWpJJ_QCD
552 prop_binSR_electron2017_bin2
553 prop_binCRF_electron2016M_bin7
554 prop_binSR_muon2018_bin6_Triboson
555 prop_binCROS_electron2018_bin17_Triboson
556 prop_binCRF_muon2016M_bin3
557 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
558 prop_binCRF_electron2017_bin2_Fake
559 prop_binSR_muon2018_bin6_TVX
560 prop_binSR_muon2016M_bin13_TVX
561 prop_binSR_electron2016M_bin11_Triboson
562 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
563 prop_binCROS_muon2016M_bin16_TTTo2L2Nu
564 prop_binCRF_muon2018_bin5_TTTo2L2Nu
565 prop_binSR_electron2016M_bin8_WrongSign
566 prop_binCRF_electron2017_bin7
567 prop_binCRF_muon2016M_bin6
568 prop_binCRTT_muon2016M_bin17_TVX
569 prop_binCROS_electron2016M_bin14_WrongSign
570 prop_binCRF_muon2016M_bin12_ZZtoLep
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

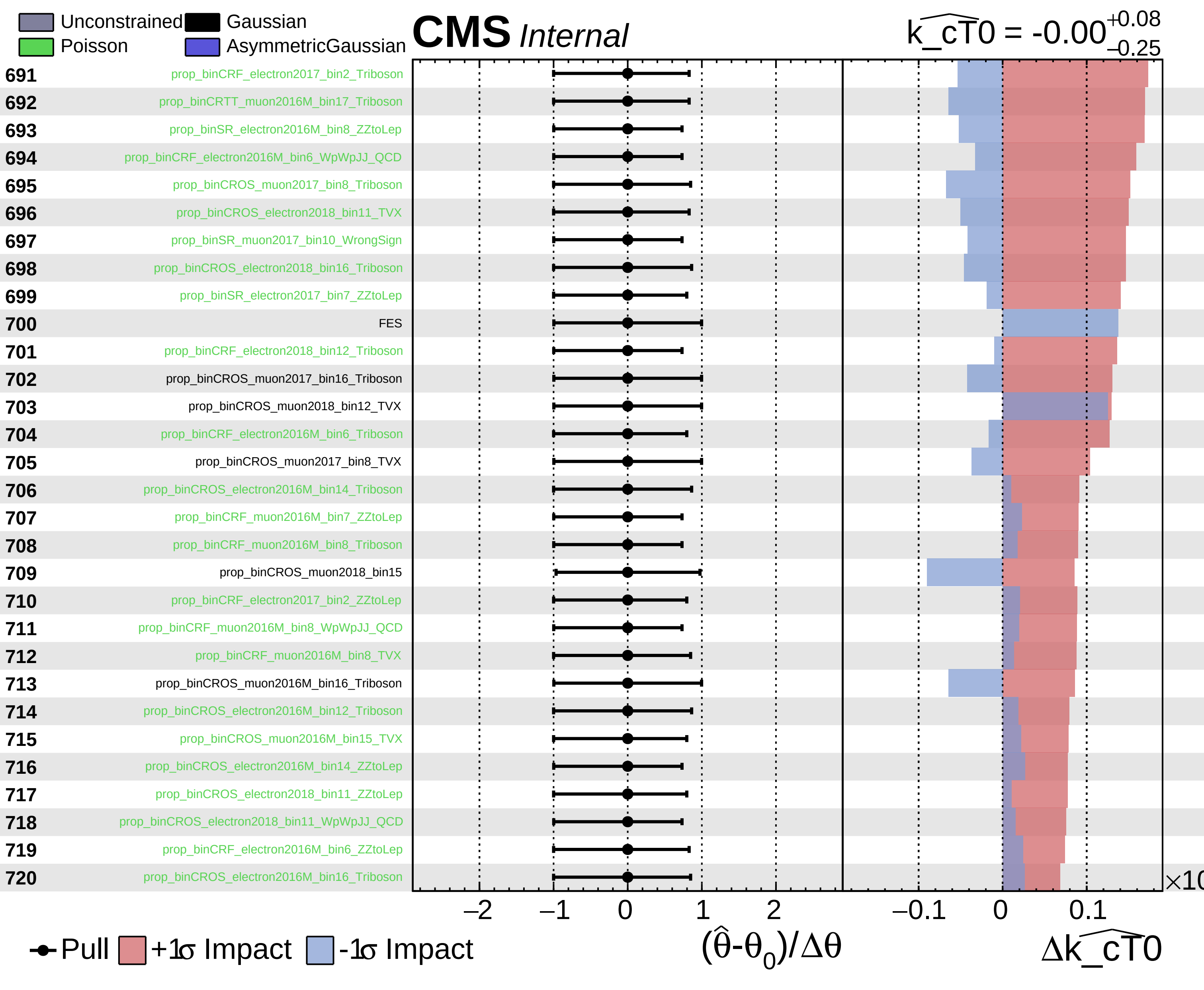

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
571 prop_binCROS_muon2017_bin16_TTTo2L2Nu
572 prop_binCRF_electron2016M_bin10
573 prop_binCRF_muon2018_bin8_WpWpJJ_QCD
574 prop_binCROS_muon2018_bin12_TTTo2L2Nu
575 prop_binCROS_muon2017_bin16_TVX
576 prop_binCRF_muon2016M_bin13
577 prop_binCRF_electron2018_bin12_WrongSign
578 prop_binSR_muon2017_bin4_TVX
579 prop_binCROS_muon2016M_bin14_TTTo2L2Nu
580 prop_binSR_electron2017_bin14_ZZtoLep
581 prop_binCROS_electron2018_bin11_TTTo2L2Nu
582 prop_binCRF_electron2018_bin3
583 prop_binCROS_electron2018_bin14_TTTo2L2Nu
584 prop_binCROS_muon2016M_bin12
585 prop_binCROS_muon2018_bin5_WZ
586 prop_binCRF_muon2016M_bin11
587 prop_binSR_electron2018_bin11_Triboson
588 prop_binSR_muon2016M_bin10_Triboson
589 prop_binCRF_muon2016M_bin8_sm
590 prop_binCROS_muon2018_bin17_ZZtoLep
591 prop_binSR_electron2017_bin13_TVX
592 prop_binCRF_electron2017_bin5
593 prop_binCRF_electron2016M_bin9
594 prop_binCRF_electron2016M_bin2
595 prop_binCRF_muon2016M_bin8_TTTo2L2Nu
596 prop_binCRF_electron2016M_bin13_TVX
597 prop_binCROS_electron2017_bin8
598 prop_binSR_electron2018_bin11_WpWpJJ_QCD
599 prop_binCROS_muon2018_bin5_TVX
600 prop_binCROS_electron2018_bin16_TTTo2L2Nu
−2 −1 0 1 2
−2 0 2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

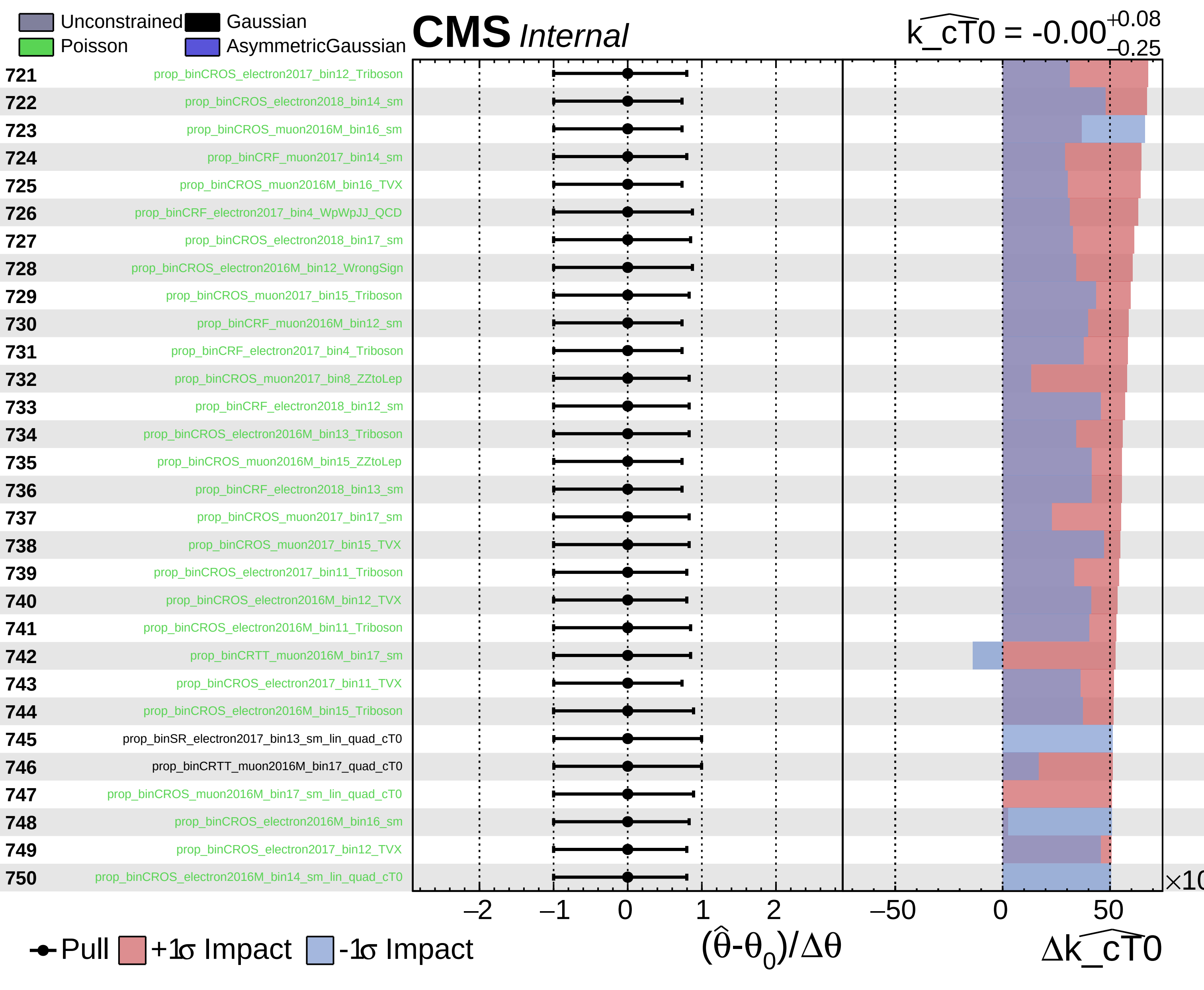

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
601 prop_binCRF_muon2017_bin8_WpWpJJ_QCD
602 prop_binCRF_muon2016M_bin14
603 prop_binSR_muon2016M_bin10_TVX
604 prop_binCRF_electron2017_bin6
605 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
606 prop_binSR_electron2017_bin11_Triboson
607 prop_binCROS_electron2016M_bin9
608 prop_binCRF_electron2016M_bin8
609 prop_binSR_electron2017_bin13_ZZtoLep
610 prop_binCRF_muon2017_bin16
611 prop_binCROS_muon2016M_bin9
612 prop_binCROS_muon2018_bin17_TVX
613 prop_binSR_muon2017_bin4_Triboson
614 prop_binSR_electron2016M_bin8_Triboson
615 prop_binCRF_electron2018_bin9
616 prop_binSR_electron2017_bin7_TVX
617 prop_binCRF_electron2018_bin12_TVX
618 prop_binCRF_muon2017_bin2
619 prop_binCRF_electron2017_bin3
620 prop_binSR_electron2016M_bin12_ZZtoLep
621 prop_binSR_muon2016M_bin12_TVX
622 prop_binCRF_electron2016M_bin3
623 prop_binCROS_electron2017_bin17_WrongSign
624 prop_binSR_electron2017_bin7_WpWpJJ_QCD
625 prop_binCRF_electron2017_bin2_TVX
626 prop_binCRF_muon2016M_bin7_WrongSign
627 prop_binSR_muon2018_bin12_ZZtoLep
628 prop_binCROS_muon2018_bin17_WrongSign
629 prop_binSR_electron2017_bin9_TVX
630 prop_binCROS_electron2016M_bin17_sm
−2 −1 0 1 2
−1 0 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$



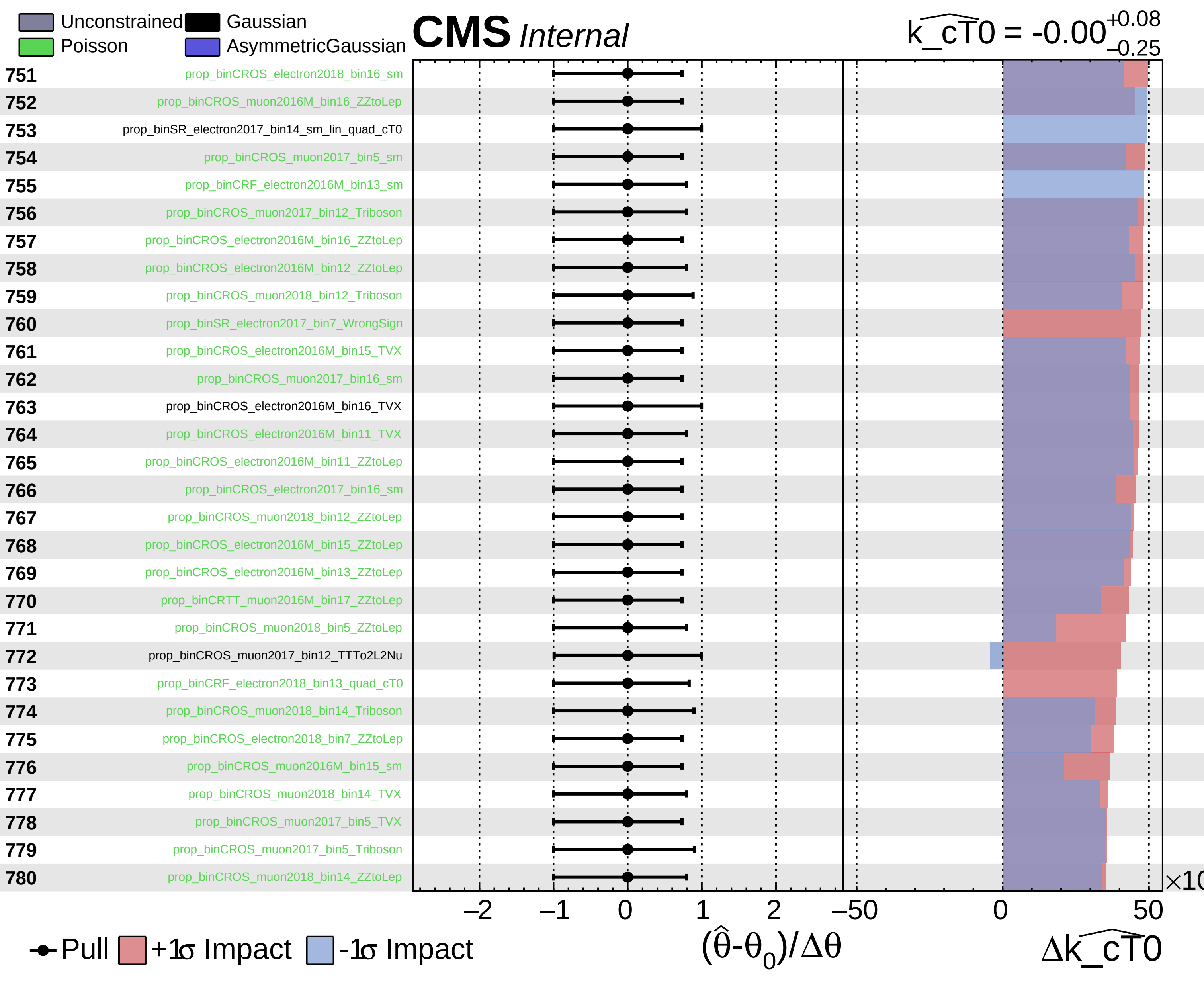
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
631 prop_binCRF_muon2018_bin5_TVX
632 prop_binCROS_electron2016M_bin14_Fake
633 prop_binSR_electron2016M_bin16_WrongSign
634 prop_binCRF_muon2018_bin8_ZZtoLep
635 prop_binCROS_electron2016M_bin16_TTTo2L2Nu
636 prop_binCRF_muon2016M_bin7_WpWpJJ_QCD
637 prop_binSR_electron2017_bin7_Triboson
638 prop_binCRF_muon2016M_bin7_TVX
639 prop_binSR_muon2018_bin6_WrongSign
640 prop_binCRF_electron2017_bin15
641 prop_binCRF_muon2018_bin5_Triboson
642 prop_binCRF_muon2016M_bin8_ZZtoLep
643 prop_binSR_muon2016M_bin13_ZZtoLep
644 prop_binCRF_muon2018_bin5_WpWpJJ_QCD
645 prop_binCROS_muon2017_bin17_Triboson
646 prop_binCRF_electron2016M_bin11
647 prop_binCROS_muon2018_bin16_Triboson
648 prop_binCRF_muon2018_bin7_ZZtoLep
649 prop_binSR_electron2016M_bin11_ZZtoLep
650 prop_binCROS_electron2018_bin11_Triboson
651 prop_binCRF_electron2018_bin10
652 prop_binCROS_electron2018_bin7_TVX
653 prop_binCROS_electron2018_bin17_WrongSign
654 prop_binCRF_muon2018_bin7_WpWpJJ_QCD
655 prop_binCRF_muon2017_bin8_ZZtoLep
656 prop_binCRF_electron2017_bin2_TTTo2L2Nu
657 prop_binCRF_muon2017_bin13
658 prop_binCROS_muon2017_bin12_TVX
659 prop_binSR_muon2018_bin6_ZZtoLep
660 prop_binCRF_muon2016M_bin7_Triboson
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
661 prop_binCROS_electron2018_bin16_TVX
662 prop_binCRF_electron2016M_bin15_sm
663 prop_binCRF_electron2017_bin2_WrongSign
664 prop_binCROS_muon2017_bin15_WrongSign
665 prop_binCRF_electron2018_bin13_WrongSign
666 prop_binCROS_electron2018_bin14_TVX
667 prop_binCRF_muon2018_bin8_Triboson
668 prop_binCRF_electron2017_bin4_TVX
669 prop_binCROS_electron2017_bin16_WrongSign
670 prop_binCROS_electron2017_bin16_Triboson
671 prop_binCROS_muon2016M_bin14_Triboson
672 prop_binSR_muon2016M_bin8_ZZtoLep
673 prop_binCROS_muon2018_bin17_sm
674 prop_binSR_muon2016M_bin9_ZZtoLep
675 prop_binCRF_electron2017_bin2_WpWpJJ_QCD
676 prop_binCRF_muon2017_bin10
677 prop_binSR_muon2017_bin13_WrongSign
678 prop_binCRF_electron2016M_bin13_Triboson
679 prop_binCROS_electron2018_bin14_Triboson
680 prop_binCROS_muon2016M_bin17_TVX
681 prop_binCRF_electron2016M_bin6_TVX
682 prop_binSR_electron2016M_bin10_WrongSign
683 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
684 prop_binCROS_electron2017_bin17_sm
685 prop_binCRF_electron2017_bin11_TVX
686 prop_binCROS_muon2018_bin16_TVX
687 prop_binSR_muon2016M_bin10_ZZtoLep
688 prop_binCRF_electron2017_bin4_ZZtoLep
689 prop_binCROS_electron2016M_bin13_sm
690 prop_binCROS_muon2016M_bin15_Triboson
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT0

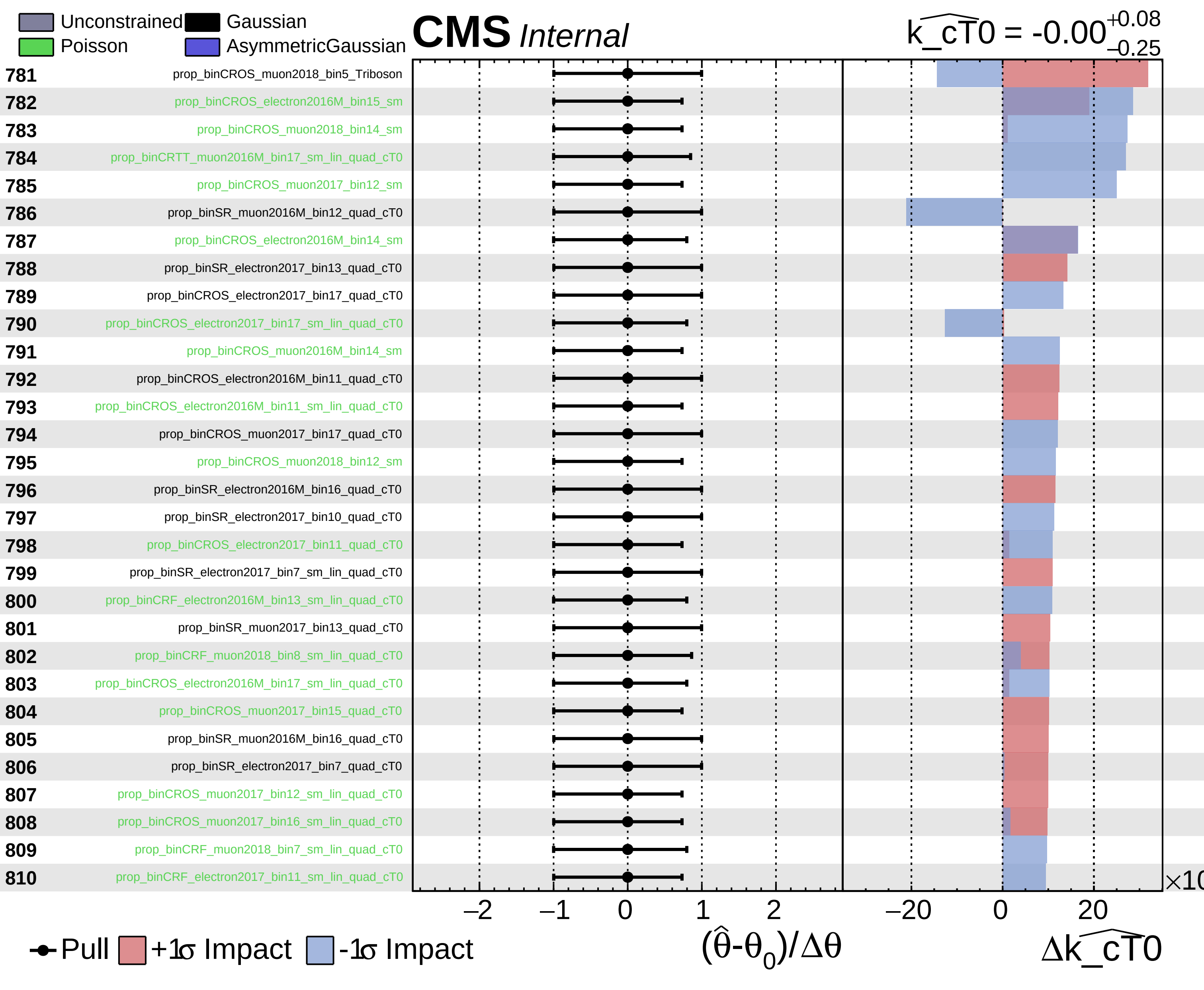

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
691 prop_binCRF_electron2017_bin2_Triboson
692 prop_binCRTT_muon2016M_bin17_Triboson
693 prop_binSR_electron2016M_bin8_ZZtoLep
694 prop_binCRF_electron2016M_bin6_WpWpJJ_QCD
695 prop_binCROS_muon2017_bin8_Triboson
696 prop_binCROS_electron2018_bin11_TVX
697 prop_binSR_muon2017_bin10_WrongSign
698 prop_binCROS_electron2018_bin16_Triboson
699 prop_binSR_electron2017_bin7_ZZtoLep
700 FES
701 prop_binCRF_electron2018_bin12_Triboson
702 prop_binCROS_muon2017_bin16_Triboson
703 prop_binCROS_muon2018_bin12_TVX
704 prop_binCRF_electron2016M_bin6_Triboson
705 prop_binCROS_muon2017_bin8_TVX
706 prop_binCROS_electron2016M_bin14_Triboson
707 prop_binCRF_muon2016M_bin7_ZZtoLep
708 prop_binCRF_muon2016M_bin8_Triboson
709 prop_binCROS_muon2018_bin15
710 prop_binCRF_electron2017_bin2_ZZtoLep
711 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
712 prop_binCRF_muon2016M_bin8_TVX
713 prop_binCROS_muon2016M_bin16_Triboson
714 prop_binCROS_electron2016M_bin12_Triboson
715 prop_binCROS_muon2016M_bin15_TVX
716 prop_binCROS_electron2016M_bin14_ZZtoLep
717 prop_binCROS_electron2018_bin11_ZZtoLep
718 prop_binCROS_electron2018_bin11_WpWpJJ_QCD
719 prop_binCRF_electron2016M_bin6_ZZtoLep
720 prop_binCROS_electron2016M_bin16_Triboson
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0

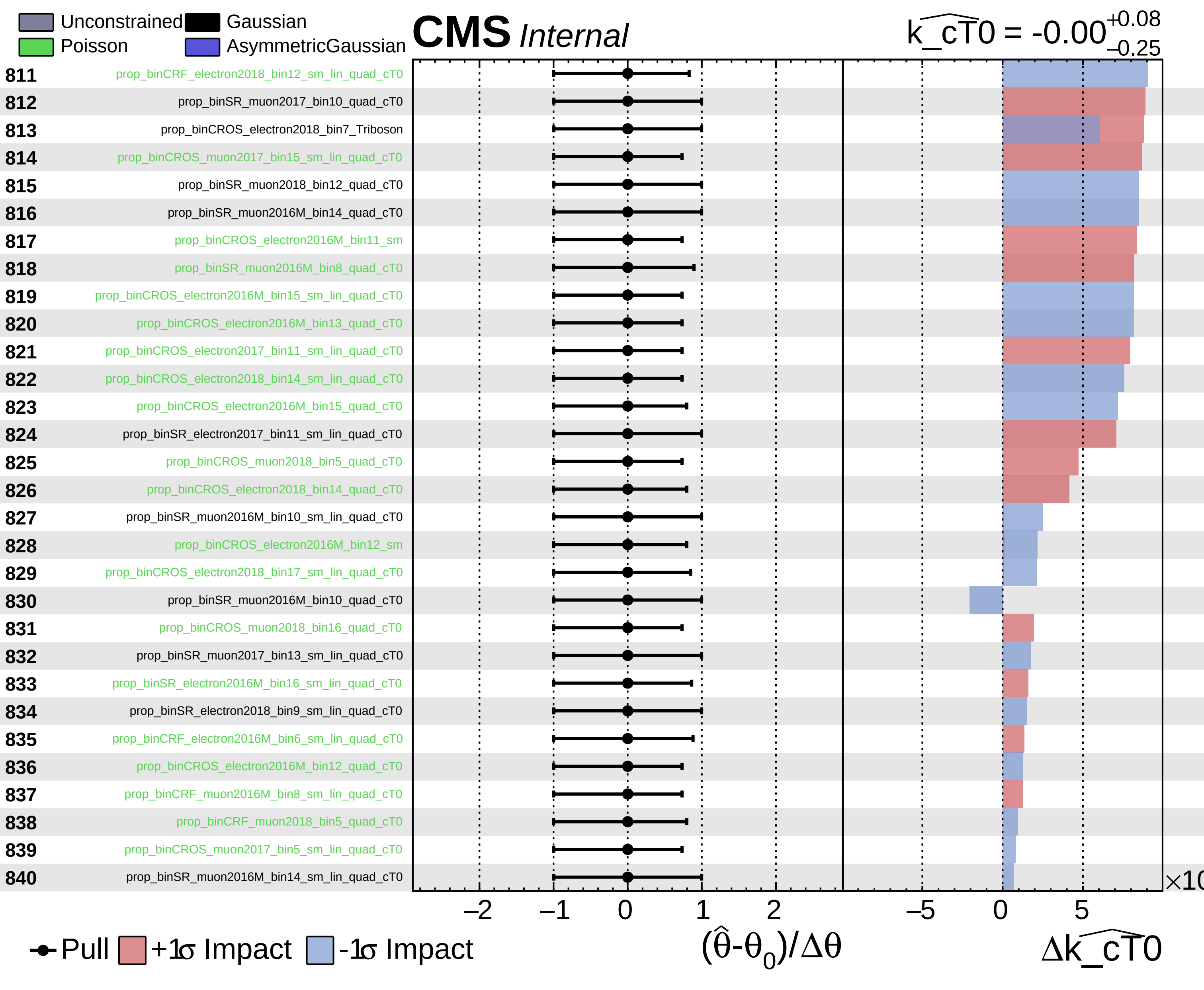


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
721 prop_binCROS_electron2017_bin12_Triboson
722 prop_binCROS_electron2018_bin14_sm
723 prop_binCROS_muon2016M_bin16_sm
724 prop_binCRF_muon2017_bin14_sm
725 prop_binCROS_muon2016M_bin16_TVX
726 prop_binCRF_electron2017_bin4_WpWpJJ_QCD
727 prop_binCROS_electron2018_bin17_sm
728 prop_binCROS_electron2016M_bin12_WrongSign
729 prop_binCROS_muon2017_bin15_Triboson
730 prop_binCRF_muon2016M_bin12_sm
731 prop_binCRF_electron2017_bin4_Triboson
732 prop_binCROS_muon2017_bin8_ZZtoLep
733 prop_binCRF_electron2018_bin12_sm
734 prop_binCROS_electron2016M_bin13_Triboson
735 prop_binCROS_muon2016M_bin15_ZZtoLep
736 prop_binCRF_electron2018_bin13_sm
737 prop_binCROS_muon2017_bin17_sm
738 prop_binCROS_muon2017_bin15_TVX
739 prop_binCROS_electron2017_bin11_Triboson
740 prop_binCROS_electron2016M_bin12_TVX
741 prop_binCROS_electron2016M_bin11_Triboson
742 prop_binCRTT_muon2016M_bin17_sm
743 prop_binCROS_electron2017_bin11_TVX
744 prop_binCROS_electron2016M_bin15_Triboson
745 prop_binSR_electron2017_bin13_sm_lin_quad_cT0
746 prop_binCRTT_muon2016M_bin17_quad_cT0
747 prop_binCROS_muon2016M_bin17_sm_lin_quad_cT0
748 prop_binCROS_electron2016M_bin16_sm
749 prop_binCROS_electron2017_bin12_TVX
750 prop_binCROS_electron2016M_bin14_sm_lin_quad_cT0
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

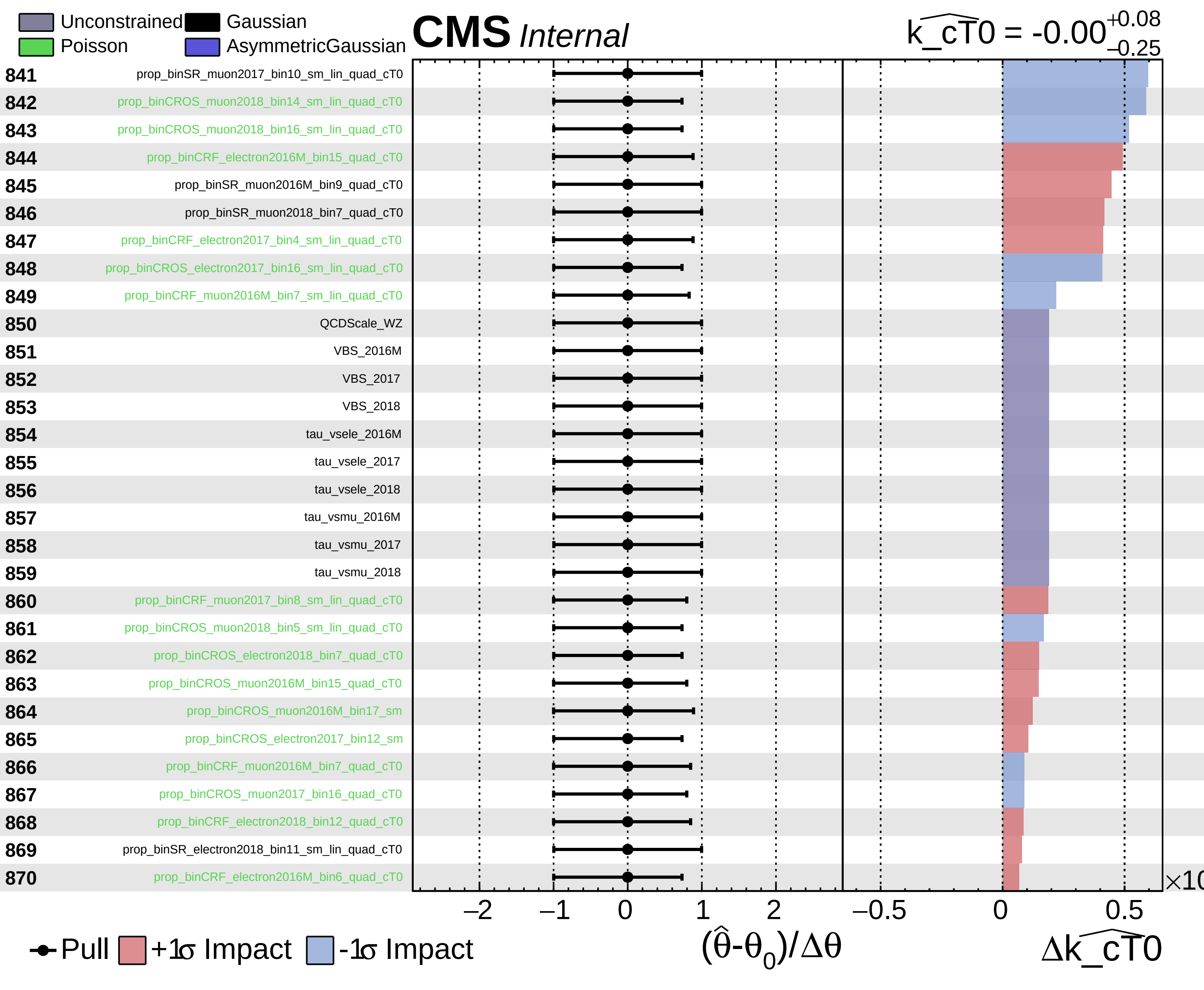


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
751 prop_binCROS_electron2018_bin16_sm
752 prop_binCROS_muon2016M_bin16_ZZtoLep
753 prop_binSR_electron2017_bin14_sm_lin_quad_cT0
754 prop_binCROS_muon2017_bin5_sm
755 prop_binCRF_electron2016M_bin13_sm
756 prop_binCROS_muon2017_bin12_Triboson
757 prop_binCROS_electron2016M_bin16_ZZtoLep
758 prop_binCROS_electron2016M_bin12_ZZtoLep
759 prop_binCROS_muon2018_bin12_Triboson
760 prop_binSR_electron2017_bin7_WrongSign
761 prop_binCROS_electron2016M_bin15_TVX
762 prop_binCROS_muon2017_bin16_sm
763 prop_binCROS_electron2016M_bin16_TVX
764 prop_binCROS_electron2016M_bin11_TVX
765 prop_binCROS_electron2016M_bin11_ZZtoLep
766 prop_binCROS_electron2017_bin16_sm
767 prop_binCROS_muon2018_bin12_ZZtoLep
768 prop_binCROS_electron2016M_bin15_ZZtoLep
769 prop_binCROS_electron2016M_bin13_ZZtoLep
770 prop_binCRTT_muon2016M_bin17_ZZtoLep
771 prop_binCROS_muon2018_bin5_ZZtoLep
772 prop_binCROS_muon2017_bin12_TTTo2L2Nu
773 prop_binCRF_electron2018_bin13_quad_cT0
774 prop_binCROS_muon2018_bin14_Triboson
775 prop_binCROS_electron2018_bin7_ZZtoLep
776 prop_binCROS_muon2016M_bin15_sm
777 prop_binCROS_muon2018_bin14_TVX
778 prop_binCROS_muon2017_bin5_TVX
779 prop_binCROS_muon2017_bin5_Triboson
780 prop_binCROS_muon2018_bin14_ZZtoLep
−2 −1 0 1 2
−50 0 50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

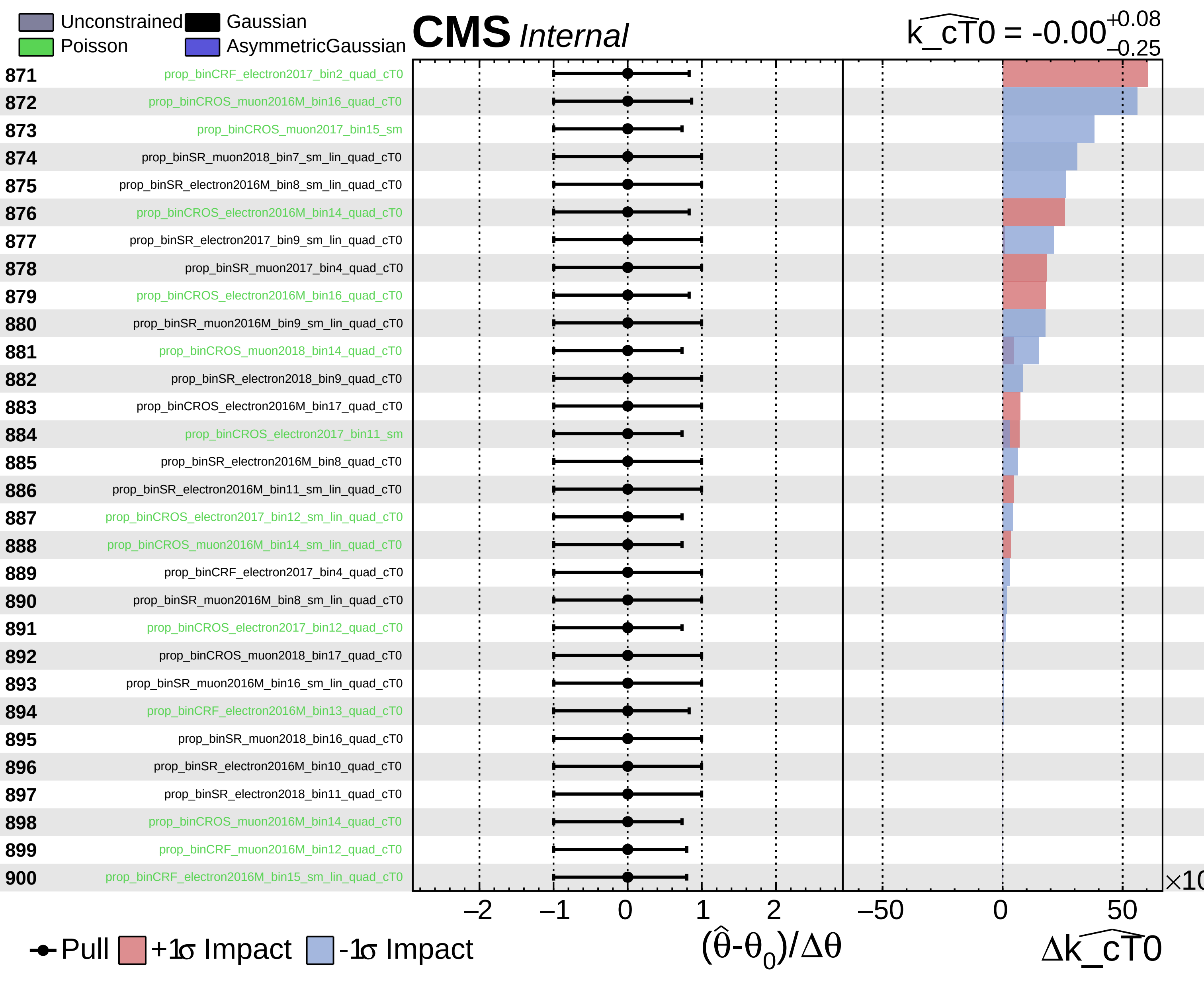

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
781 prop_binCROS_muon2018_bin5_Triboson
782 prop_binCROS_electron2016M_bin15_sm
783 prop_binCROS_muon2018_bin14_sm
784 prop_binCRTT_muon2016M_bin17_sm_lin_quad_cT0
785 prop_binCROS_muon2017_bin12_sm
786 prop_binSR_muon2016M_bin12_quad_cT0
787 prop_binCROS_electron2016M_bin14_sm
788 prop_binSR_electron2017_bin13_quad_cT0
789 prop_binCROS_electron2017_bin17_quad_cT0
790 prop_binCROS_electron2017_bin17_sm_lin_quad_cT0
791 prop_binCROS_muon2016M_bin14_sm
792 prop_binCROS_electron2016M_bin11_quad_cT0
793 prop_binCROS_electron2016M_bin11_sm_lin_quad_cT0
794 prop_binCROS_muon2017_bin17_quad_cT0
795 prop_binCROS_muon2018_bin12_sm
796 prop_binSR_electron2016M_bin16_quad_cT0
797 prop_binSR_electron2017_bin10_quad_cT0
798 prop_binCROS_electron2017_bin11_quad_cT0
799 prop_binSR_electron2017_bin7_sm_lin_quad_cT0
800 prop_binCRF_electron2016M_bin13_sm_lin_quad_cT0
801 prop_binSR_muon2017_bin13_quad_cT0
802 prop_binCRF_muon2018_bin8_sm_lin_quad_cT0
803 prop_binCROS_electron2016M_bin17_sm_lin_quad_cT0
804 prop_binCROS_muon2017_bin15_quad_cT0
805 prop_binSR_muon2016M_bin16_quad_cT0
806 prop_binSR_electron2017_bin7_quad_cT0
807 prop_binCROS_muon2017_bin12_sm_lin_quad_cT0
808 prop_binCROS_muon2017_bin16_sm_lin_quad_cT0
809 prop_binCRF_muon2018_bin7_sm_lin_quad_cT0
810 prop_binCRF_electron2017_bin11_sm_lin_quad_cT0
−2 −1 0 1 2
−20 0 20
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0

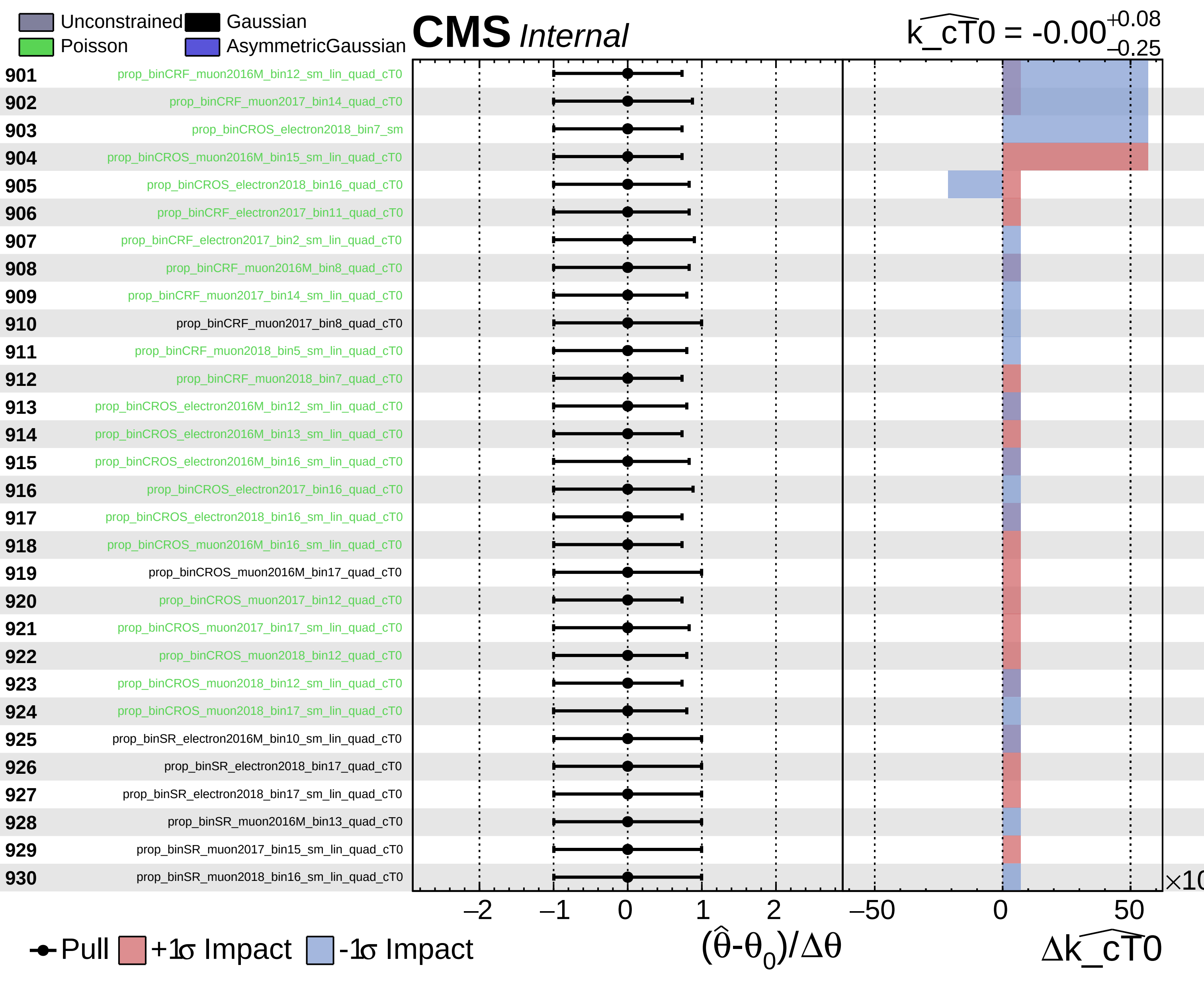


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0}$ = $-0.00^{+0.08}_{-0.25}$
811 prop_binCRF_electron2018_bin12_sm_lin_quad_cT0
812 prop_binSR_muon2017_bin10_quad_cT0
813 prop_binCROS_electron2018_bin7_Triboson
814 prop_binCROS_muon2017_bin15_sm_lin_quad_cT0
815 prop_binSR_muon2018_bin12_quad_cT0
816 prop_binSR_muon2016M_bin14_quad_cT0
817 prop_binCROS_electron2016M_bin11_sm
818 prop_binSR_muon2016M_bin8_quad_cT0
819 prop_binCROS_electron2016M_bin15_sm_lin_quad_cT0
820 prop_binCROS_electron2016M_bin13_quad_cT0
821 prop_binCROS_electron2017_bin11_sm_lin_quad_cT0
822 prop_binCROS_electron2018_bin14_sm_lin_quad_cT0
823 prop_binCROS_electron2016M_bin15_quad_cT0
824 prop_binSR_electron2017_bin11_sm_lin_quad_cT0
825 prop_binCROS_muon2018_bin5_quad_cT0
826 prop_binCROS_electron2018_bin14_quad_cT0
827 prop_binSR_muon2016M_bin10_sm_lin_quad_cT0
828 prop_binCROS_electron2016M_bin12_sm
829 prop_binCROS_electron2018_bin17_sm_lin_quad_cT0
830 prop_binSR_muon2016M_bin10_quad_cT0
831 prop_binCROS_muon2018_bin16_quad_cT0
832 prop_binSR_muon2017_bin13_sm_lin_quad_cT0
833 prop_binSR_electron2016M_bin16_sm_lin_quad_cT0
834 prop_binSR_electron2018_bin9_sm_lin_quad_cT0
835 prop_binCRF_electron2016M_bin6_sm_lin_quad_cT0
836 prop_binCROS_electron2016M_bin12_quad_cT0
837 prop_binCRF_muon2016M_bin8_sm_lin_quad_cT0
838 prop_binCRF_muon2018_bin5_quad_cT0
839 prop_binCROS_muon2017_bin5_sm_lin_quad_cT0
840 prop_binSR_muon2016M_bin14_sm_lin_quad_cT0
−2 −1 0 1 2
−5 0 5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$

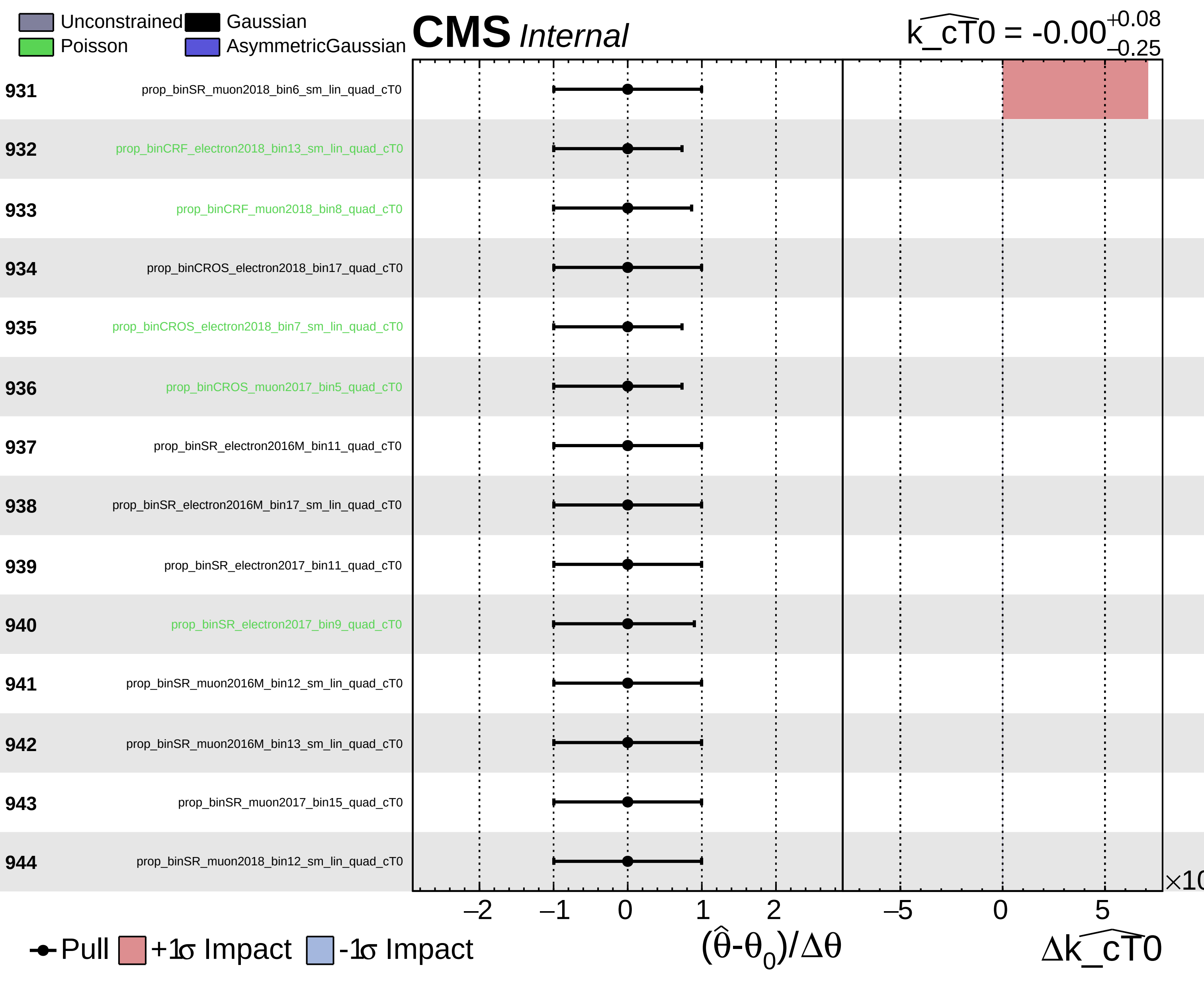


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT0} = -0.00^{+0.08}_{-0.25}$
841 prop_binSR_muon2017_bin10_sm_lin_quad_cT0
842 prop_binCROS_muon2018_bin14_sm_lin_quad_cT0
843 prop_binCROS_muon2018_bin16_sm_lin_quad_cT0
844 prop_binCRF_electron2016M_bin15_quad_cT0
845 prop_binSR_muon2016M_bin9_quad_cT0
846 prop_binSR_muon2018_bin7_quad_cT0
847 prop_binCRF_electron2017_bin4_sm_lin_quad_cT0
848 prop_binCROS_electron2017_bin16_sm_lin_quad_cT0
849 prop_binCRF_muon2016M_bin7_sm_lin_quad_cT0
850 QCDScale_WZ
851 VBS_2016M
852 VBS_2017
853 VBS_2018
854 tau_vsele_2016M
855 tau_vsele_2017
856 tau_vsele_2018
857 tau_vsmu_2016M
858 tau_vsmu_2017
859 tau_vsmu_2018
860 prop_binCRF_muon2017_bin8_sm_lin_quad_cT0
861 prop_binCROS_muon2018_bin5_sm_lin_quad_cT0
862 prop_binCROS_electron2018_bin7_quad_cT0
863 prop_binCROS_muon2016M_bin15_quad_cT0
864 prop_binCROS_muon2016M_bin17_sm
865 prop_binCROS_electron2017_bin12_sm
866 prop_binCRF_muon2016M_bin7_quad_cT0
867 prop_binCROS_muon2017_bin16_quad_cT0
868 prop_binCRF_electron2018_bin12_quad_cT0
869 prop_binSR_electron2018_bin11_sm_lin_quad_cT0
870 prop_binCRF_electron2016M_bin6_quad_cT0
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
871 prop_binCRF_electron2017_bin2_quad_cT0
872 prop_binCROS_muon2016M_bin16_quad_cT0
873 prop_binCROS_muon2017_bin15_sm
874 prop_binSR_muon2018_bin7_sm_lin_quad_cT0
875 prop_binSR_electron2016M_bin8_sm_lin_quad_cT0
876 prop_binCROS_electron2016M_bin14_quad_cT0
877 prop_binSR_electron2017_bin9_sm_lin_quad_cT0
878 prop_binSR_muon2017_bin4_quad_cT0
879 prop_binCROS_electron2016M_bin16_quad_cT0
880 prop_binSR_muon2016M_bin9_sm_lin_quad_cT0
881 prop_binCROS_muon2018_bin14_quad_cT0
882 prop_binSR_electron2018_bin9_quad_cT0
883 prop_binCROS_electron2016M_bin17_quad_cT0
884 prop_binCROS_electron2017_bin11_sm
885 prop_binSR_electron2016M_bin8_quad_cT0
886 prop_binSR_electron2016M_bin11_sm_lin_quad_cT0
887 prop_binCROS_electron2017_bin12_sm_lin_quad_cT0
888 prop_binCROS_muon2016M_bin14_sm_lin_quad_cT0
889 prop_binCRF_electron2017_bin4_quad_cT0
890 prop_binSR_muon2016M_bin8_sm_lin_quad_cT0
891 prop_binCROS_electron2017_bin12_quad_cT0
892 prop_binCROS_muon2018_bin17_quad_cT0
893 prop_binSR_muon2016M_bin16_sm_lin_quad_cT0
894 prop_binCRF_electron2016M_bin13_quad_cT0
895 prop_binSR_muon2018_bin16_quad_cT0
896 prop_binSR_electron2016M_bin10_quad_cT0
897 prop_binSR_electron2018_bin11_quad_cT0
898 prop_binCROS_muon2016M_bin14_quad_cT0
899 prop_binCRF_muon2016M_bin12_quad_cT0
900 prop_binCRF_electron2016M_bin15_sm_lin_quad_cT0
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cT0



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
901 prop_binCRF_muon2016M_bin12_sm_lin_quad_cT0
902 prop_binCRF_muon2017_bin14_quad_cT0
903 prop_binCROS_electron2018_bin7_sm
904 prop_binCROS_muon2016M_bin15_sm_lin_quad_cT0
905 prop_binCROS_electron2018_bin16_quad_cT0
906 prop_binCRF_electron2017_bin11_quad_cT0
907 prop_binCRF_electron2017_bin2_sm_lin_quad_cT0
908 prop_binCRF_muon2016M_bin8_quad_cT0
909 prop_binCRF_muon2017_bin14_sm_lin_quad_cT0
910 prop_binCRF_muon2017_bin8_quad_cT0
911 prop_binCRF_muon2018_bin5_sm_lin_quad_cT0
912 prop_binCRF_muon2018_bin7_quad_cT0
913 prop_binCROS_electron2016M_bin12_sm_lin_quad_cT0
914 prop_binCROS_electron2016M_bin13_sm_lin_quad_cT0
915 prop_binCROS_electron2016M_bin16_sm_lin_quad_cT0
916 prop_binCROS_electron2017_bin16_quad_cT0
917 prop_binCROS_electron2018_bin16_sm_lin_quad_cT0
918 prop_binCROS_muon2016M_bin16_sm_lin_quad_cT0
919 prop_binCROS_muon2016M_bin17_quad_cT0
920 prop_binCROS_muon2017_bin12_quad_cT0
921 prop_binCROS_muon2017_bin17_sm_lin_quad_cT0
922 prop_binCROS_muon2018_bin12_quad_cT0
923 prop_binCROS_muon2018_bin12_sm_lin_quad_cT0
924 prop_binCROS_muon2018_bin17_sm_lin_quad_cT0
925 prop_binSR_electron2016M_bin10_sm_lin_quad_cT0
926 prop_binSR_electron2018_bin17_quad_cT0
927 prop_binSR_electron2018_bin17_sm_lin_quad_cT0
928 prop_binSR_muon2016M_bin13_quad_cT0
929 prop_binSR_muon2017_bin15_sm_lin_quad_cT0
930 prop_binSR_muon2018_bin16_sm_lin_quad_cT0
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT0


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT0 = -0.00 +0.08 −0.25
931 prop_binSR_muon2018_bin6_sm_lin_quad_cT0
932 prop_binCRF_electron2018_bin13_sm_lin_quad_cT0
933 prop_binCRF_muon2018_bin8_quad_cT0
934 prop_binCROS_electron2018_bin17_quad_cT0
935 prop_binCROS_electron2018_bin7_sm_lin_quad_cT0
936 prop_binCROS_muon2017_bin5_quad_cT0
937 prop_binSR_electron2016M_bin11_quad_cT0
938 prop_binSR_electron2016M_bin17_sm_lin_quad_cT0
939 prop_binSR_electron2017_bin11_quad_cT0
940 prop_binSR_electron2017_bin9_quad_cT0
941 prop_binSR_muon2016M_bin12_sm_lin_quad_cT0
942 prop_binSR_muon2016M_bin13_sm_lin_quad_cT0
943 prop_binSR_muon2017_bin15_quad_cT0
944 prop_binSR_muon2018_bin12_sm_lin_quad_cT0
−2 −1 0 1 2
−5 0 5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT0}$